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Practical No - 1

Aim - VLAN - Inter - VLAN.

Problem statement - Design and configure a network that ~~was~~ use multiple VLANs to segment traffic, implement inter VLAN routing using a router-on-a-stick or layer-3 switch, so that hosts in different VLANs can communicate securely verify connectivity and ensure proper isolation between broadcast domains.

Theory - What is VLAN?

A VLAN is a logical Network created inside a switch. It allows you to divide one physical network into multiple smaller networks.

Access Port:

- An access port connect end devices (PC).
- It carries of only one VLAN

Trunk Port:

- A trunk port carries traffic from multiple VLAN between devices.
- It uses 8.02.1Q (dot1 Q) tagging to identify VLAN

Network topology -

PC1 → VLAN 10
PC2 → VLAN 20
PC3 → VLAN 30

- All PCs connected to switch
- switch connected to Router
- without trunk port

Step-1) Assign IP to PCs.
- PC0 → VLAN 10 → 192.168.10.1
- PC1 → VLAN 20 → 192.168.20.1
- PC2 → VLAN 30 → 192.168.30.1

Step-2) configure VLANs on switch
Go to switch CLI : - enable
- configure terminal
- VLAN 10
- name sales.
- exit.

- VLAN 20
- name accounts
- exit.

- VLAN 30
- name stocks
- exit

Step-3) Assign ports to VLANs

- PC1 on FastEthernet 0/1
- PC2 on FastEthernet 0/2
- PC3 on FastEthernet 0/3

connection of PC to switch

- interface fa 0/1
- switchport mode access
- switchport access VLAN 10
- exit.

- interface fa 0/2
- switchport mode access
- switchport access VLAN 20
- exit

- interface fa 0/3
- switchport mode access
- switchport access VLAN 30
- exit.

step-4) - Configure trunk between switch & router
Router connects on switch "fa 0/24"

- interface fa 0/24
- switchport mode trunk
- switchport trunk encapsulation dot 1q
- exit

step-5) - configure Router-on-a-stick
Router interface g0/0

- enable
- configure terminal
- interface g0/0
- no shutdown
- exit

ARISE & SHINE

- interface g0/0.10
- encapsulation dot1eQs 10
- ip address 192.168.10.1 255.255.255.0
- exit

- interface g0/0.20
- encapsulation dot1eQ 20
- ip address 192.168.20.1 255.255.255.0
- exit.

step-6) - Test connectivity.

- ping 192.168.10.1
- ping 192.168.20.1
- ping 192.168.30.1

Conclusion - VLAN help divide a single physical network into multiple logical network, improving security, performance and management. By using access ~~point~~ port, each PC is placed in its correct VLAN traffic to the router. The router user sub-interfaces to route packet between VLANs enabling inter-VLAN communication.

ARISE & SHINE

PC0

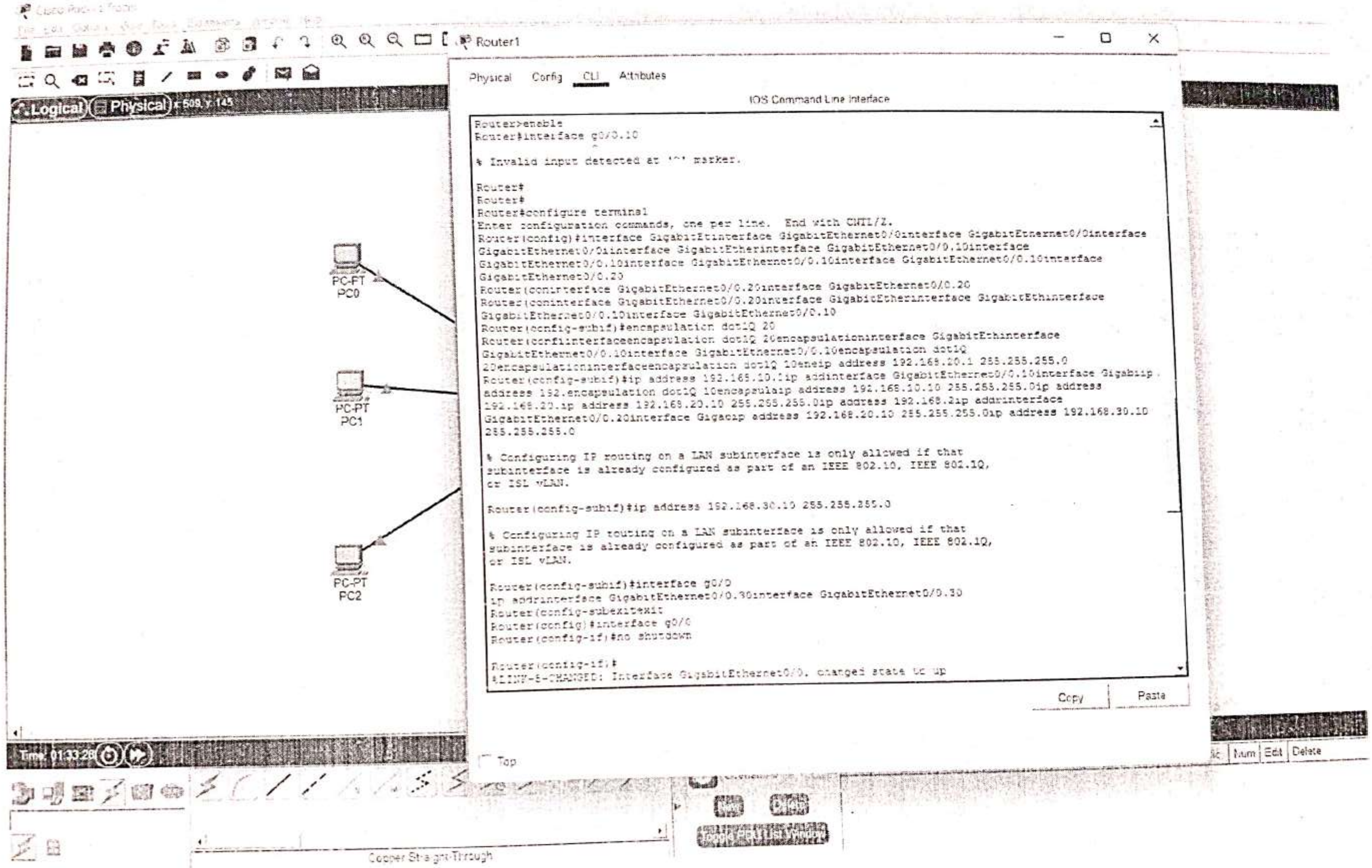
Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=4ms TTL=128
Reply from 192.168.10.10: bytes=32 time=9ms TTL=128
Reply from 192.168.10.10: bytes=32 time=9ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 5ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>
C:\>
```

Top

New Delete



Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>enable
Router#interface g0/0.10
% Invalid input detected at '^' marker.
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEtinterface GigabitEthernet0/0interface GigabitEthernet0/0interface
GigabitEthernet0/0interface GigabitEtherinterface GigabitEthernet0/0.10interface
GigabitEthernet0/0.10interface GigabitEthernet0/0.10interface GigabitEthernet0/0.10interface
GigabitEthernet0/0.20
Router(coninterface GigabitEthernet0/0.20interface GigabitEthernet0/0.20
Router(coninterface GigabitEthernet0/0.20interface GigabitEtherinterface GigabitEthinterface
GigabitEthernet0/0.10interface GigabitEthernet0/0.10
Router(config-subif)#encapsulation dot1Q 20
Router(confiinterfaceencapsulation dot1Q 20encapsulationinterface GigabitEthinterface
GigabitEthernet0/0.10interface GigabitEthernet0/0.10encapsulation dot1Q
20encapsulationinterfaceencapsulation dot1Q 10eneip address 192.168.20.1 255.255.255.0
Router(config-subif)#ip address 192.165.10.1ip addinterface GigabitEthernet0/0.10interface Gigabiip
address 192.encapsulation dot1Q 10encapsulaip address 192.168.10.10 255.255.255.0ip address
192.168.20.ip address 192.168.20.10 255.255.255.0ip address 192.168.2ip addrinterface
GigabitEthernet0/0.20interface Gigacip address 192.168.20.10 255.255.255.0ip address 192.168.30.10
255.255.255.0
% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.
Router(config-subif)#ip address 192.168.30.10 255.255.255.0
% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.
Router(config-subif)#interface g0/0
ip addrinterface GigabitEthernet0/0.30interface GigabitEthernet0/0.30
Router(config-subexitexit
Router(config)#interface g0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
Copy
Paste
Top
Logical
Physical
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Copper Straight-Through
Num
Edit
Delete

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address 192.168.10.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

Automatic Static

IPv6 Address /

Link Local Address FE80::2D0:58FF:FE54:741E

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication MD5

Username

Password

Top

Practical No-2

Aim - configuring standard and extended ACLs.

Problem statement → create and apply both standard and extended access control list (ACLs) to control. Traffic flow within. a Network implement rule to permit or deny specific port, subnets, protocals and ports, test the ACL functionality to ensures correct filtering without disrupting authorized communication.

Theory - What is an ACL (Access Control list)?
An ACL is a set of rules used in networking to allow or block traffic.

- Way do we use ACL?

ACLs are used to
- Improve network security
- control who can access what
- Block unauthorized user
- allow only needed traffic.

- Types of ACLs:

(1) Standard ACL - Standard ACL check only the source.

only the source IP address, who is sending the traffic

It does not check
- Destination
- Protocal
- Port Number

eg - allow traffic from 192.168.1.10
- Block traffic from 192.168.1.20

2) extended ACL - extended ACL is more powerful
- source IP
- destination IP
- protocal (TCP, UDP, ICMP)
- port no. (HTTP, FTP, SSH, etc).

eg - allow PC1 to access web server (port 80)
- block PC2 from using FTP (port 21)

What can ACL rules do?
ACL rules can : ① permit (allow) traffic
② deny (block) traffic.

- How ACL control traffic

① A packet enter the router
② ACL checks the packet
③ If rule matches - permit or deny
④ If no rule matches -deny by default.

- Testing ACL functionality

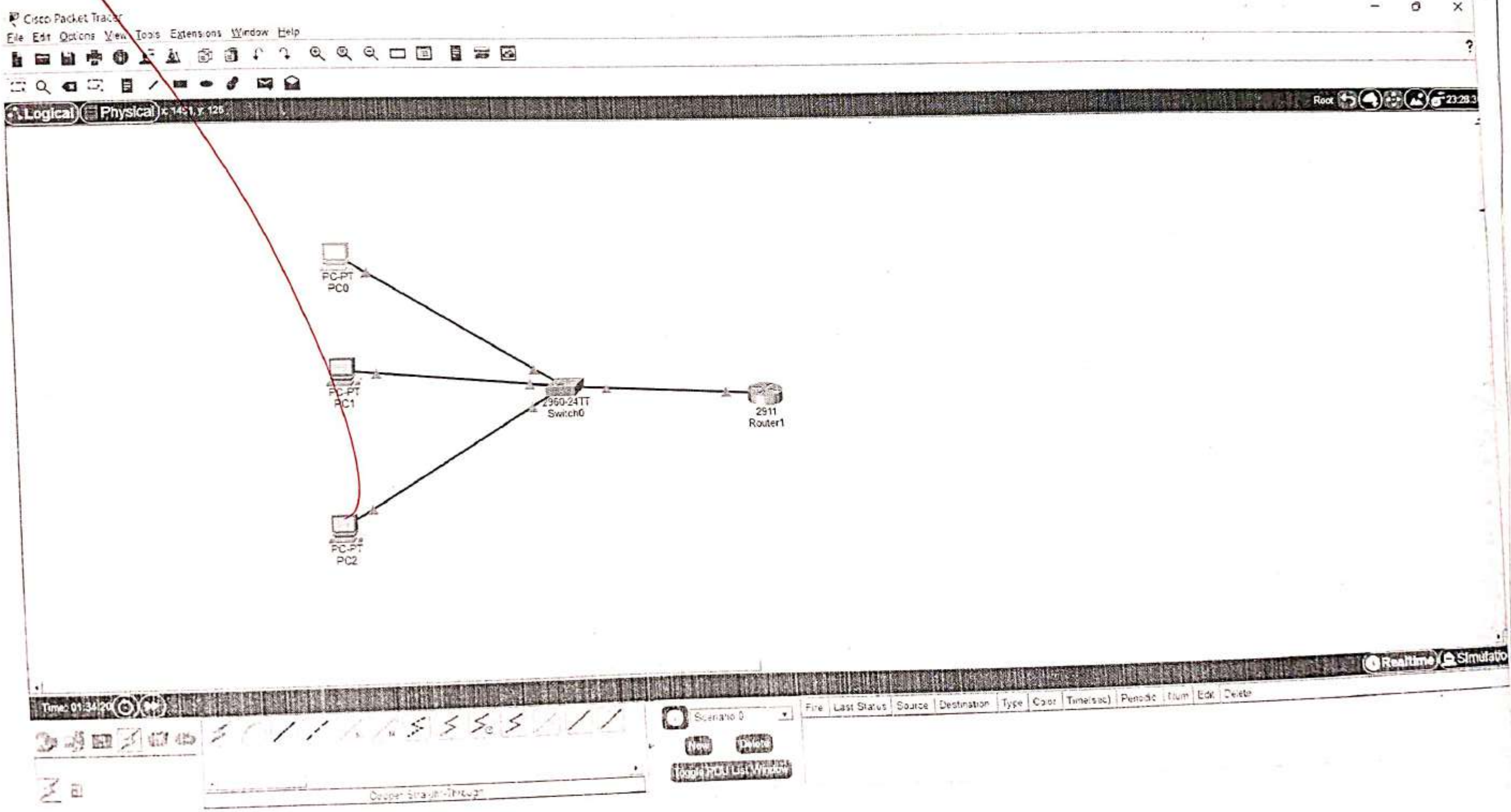
After applying ACL we test to make sure
- Allowed users can communicates
- Blocked users control access
- No important service is distributed.

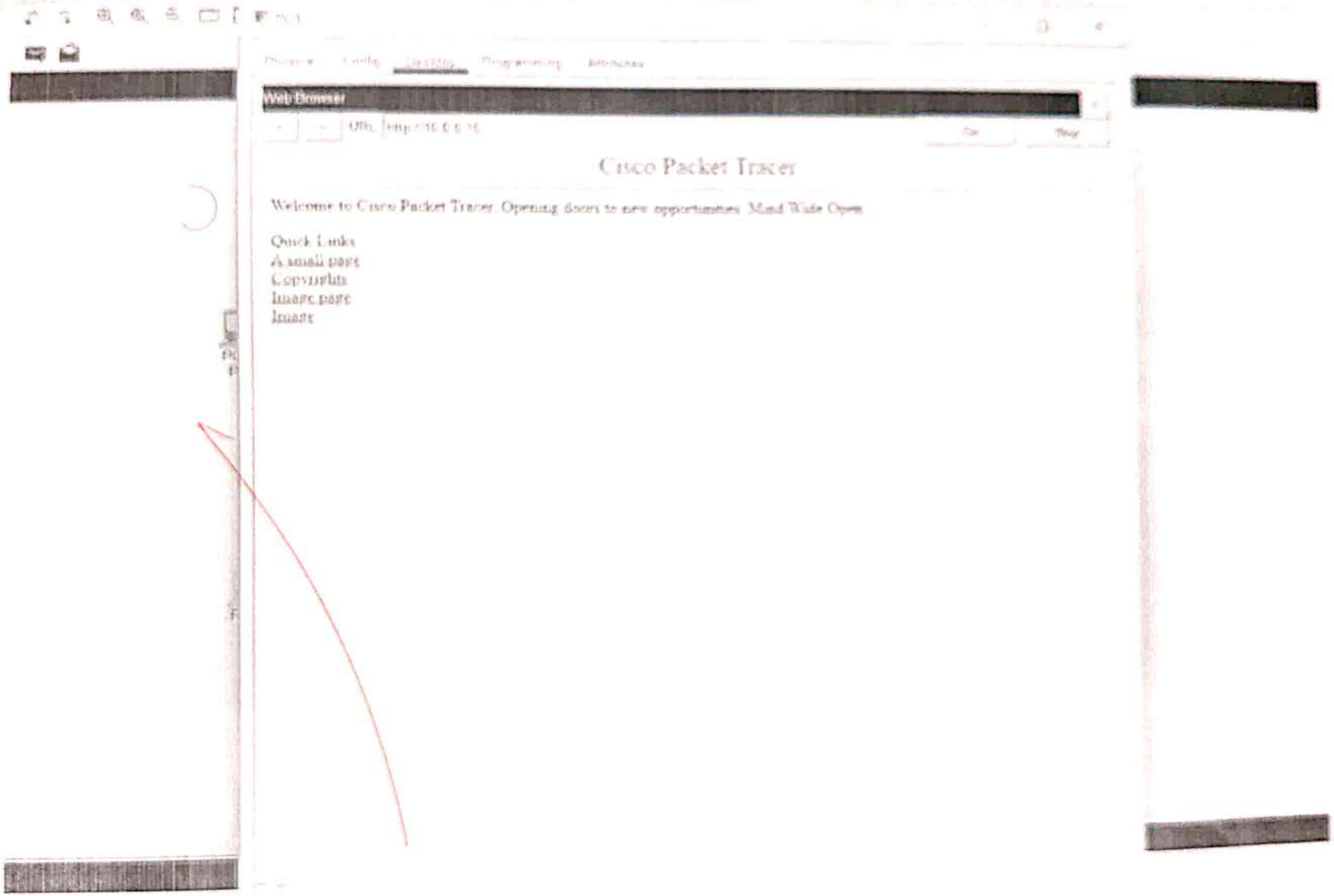
- Common testing methods

- ping (ICMP)
- web access (HTTP / HTTPS)
- FTP or SSH
- show ACL command on router

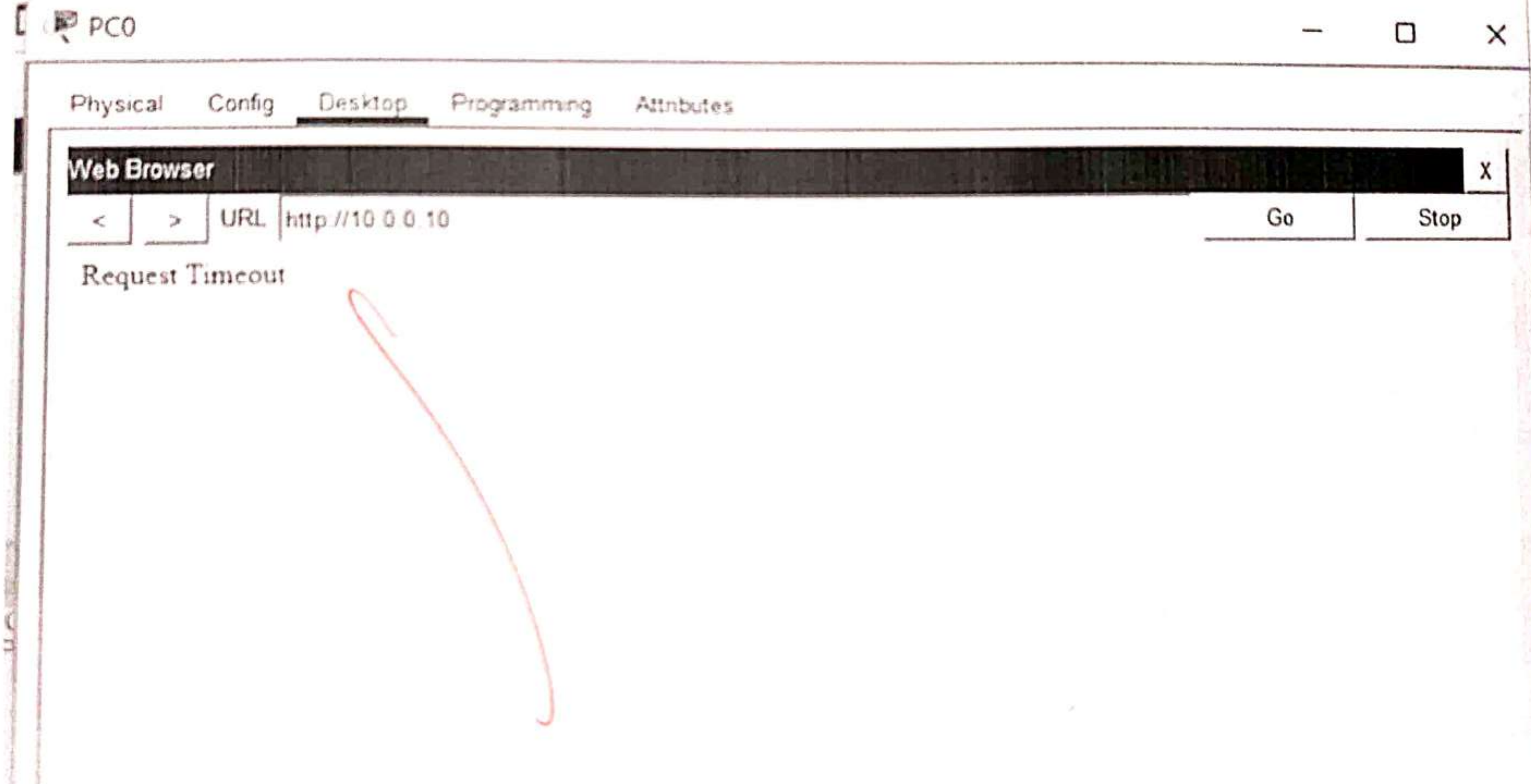
conclusion - Hence, we studied configuring standard and extended ACL.

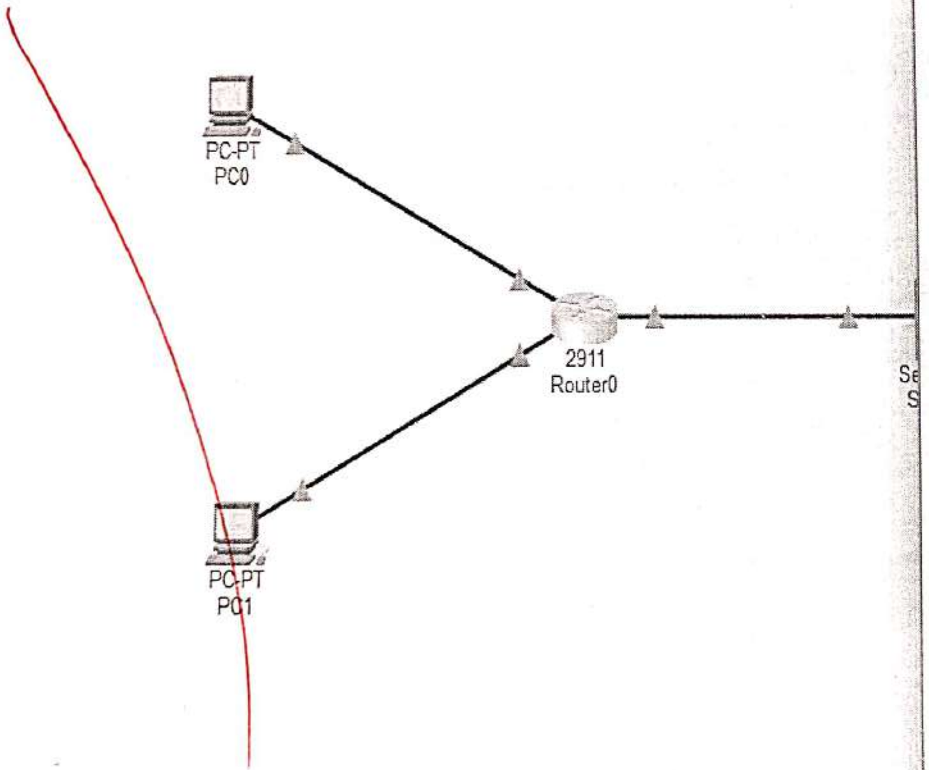
29/1/26


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1451, y: 125
Root
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
2960-24TT
Switch0
2911
Router1
Realtime
Simulatio
Time: 01:34:20
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Copper Straight-Through

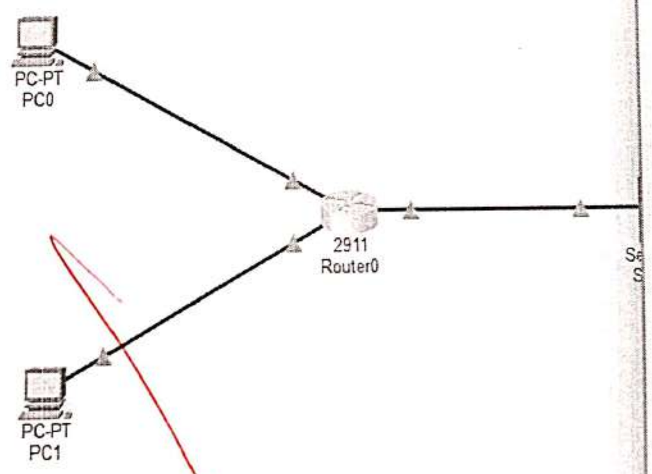


Web Browser
URL http://10.0.0.15
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links
A small page
Copyrights
Image page
Image


PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://10.0.0.10
Go
Stop
Request Timeout




```
Router#show ip interface g0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Internet address is 192.168.1.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is 100
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
 --More--   IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Probe proxy name replies are disabled
  Policy routing is disabled
  Network address translation is disabled
  BGP Policy Mapping is disabled
  Input features: MCI Check
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled


Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
```



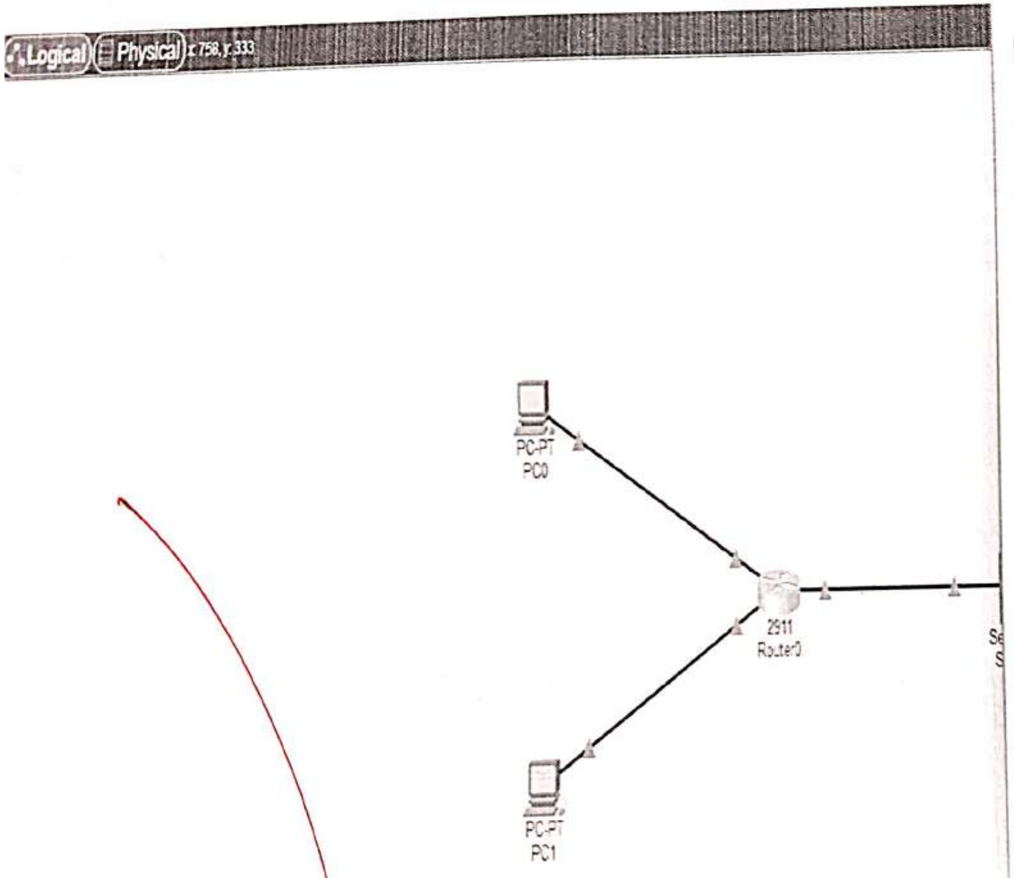

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access list 100 deny tcp 192.168.1.10 0.0.0.0 10.0.0.10 0.0.0.0 eq 80
                      ^
% Invalid input detected at '^' marker.

Router(config)#access-list 100 deny tcp 192.168.1.10 0.0.0.0 10.0.0.10 0.0.0.0 eq 80
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#exit
Router(config)#show access-list
                ^
% Invalid input detected at '^' marker.

Router(config)#Show access-list
                ^
% Invalid input detected at '^' marker.

Router(config)#Show access-lists
                ^
% Invalid input detected at '^' marker.

Router(config)#show access-lists
                ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show access-list
Extended IP access list 100
    10 deny tcp host 192.168.1.10 host 10.0.0.10 eq www
    20 permit ip any any

Router#show ip interface g0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Internet address is 192.168.1.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
```

Copy | Paste

Top

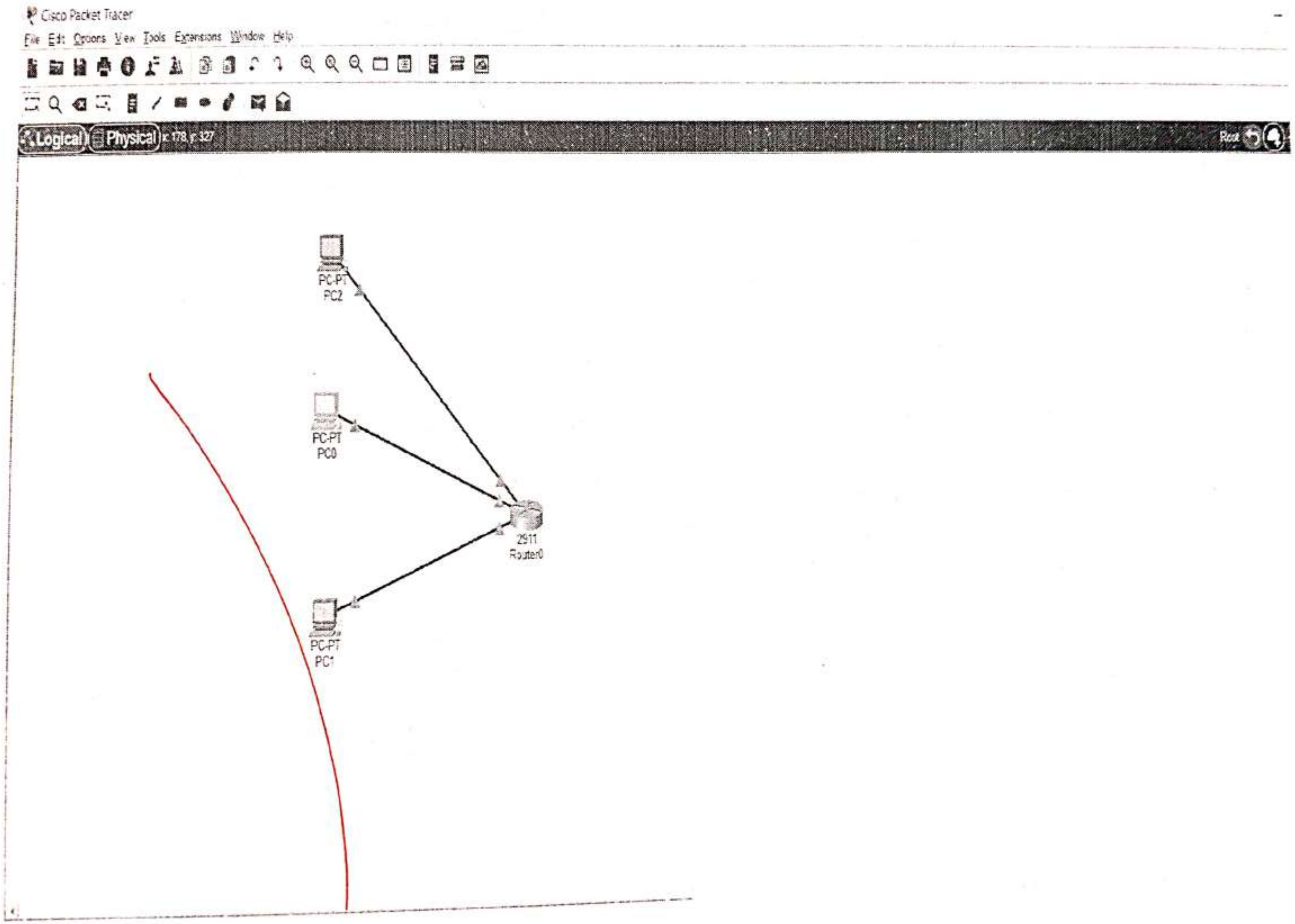
Scenario 0

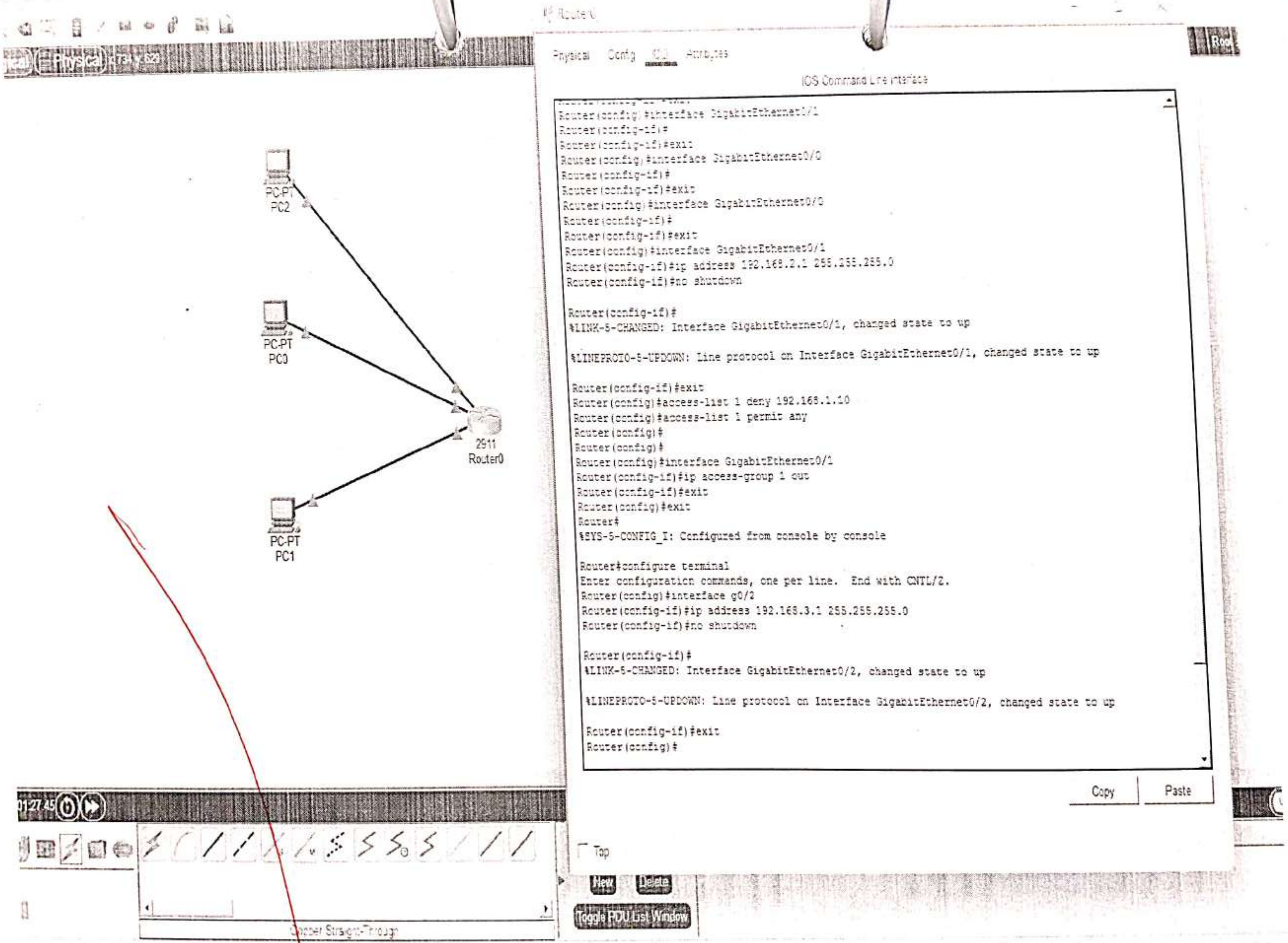
Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete

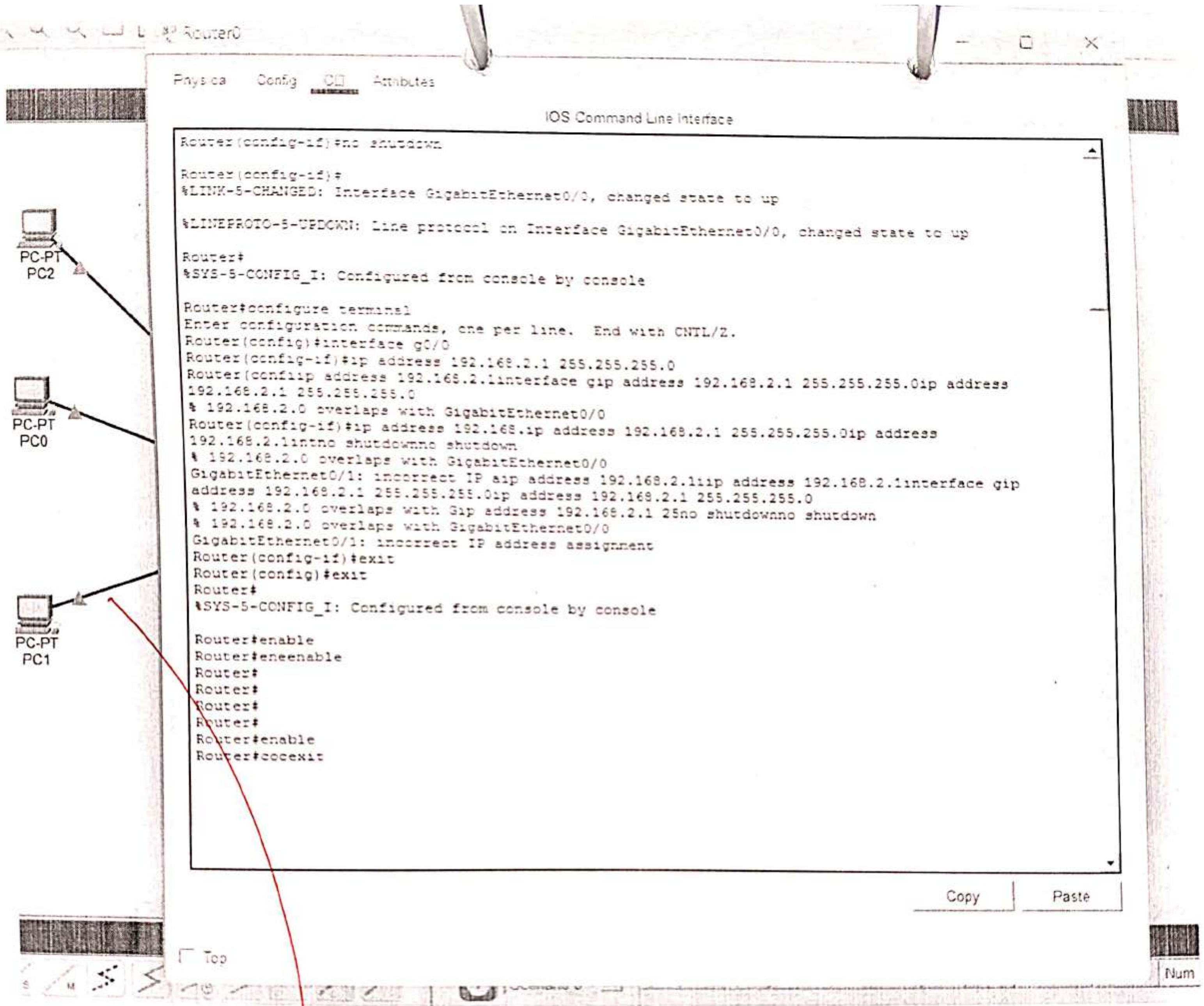


IOS Command Line Interface

```
The virtual terminal password is used to protect
access to the router over a network interface.
Enter virtual terminal password:

Press RETURN to get started!



Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface g0/2
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
```


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 178, y: 327
Root
PC-PT
PC2
PC-PT
PC0
PC-PT
PC1
2911
Router0


Physical
PC-PT
PC2
PC-PT
PC0
PC-PT
PC1
2911
Router0
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#interface GigabitEthernet1/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#access-list 1 deny 192.168.1.10
Router(config)#access-list 1 permit any
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip access-group 1 out
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/2
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
Router(config-if)#exit
Router(config)#
Copy
Paste
Top
New
Delete
Toggle PDU List Window
Copper Straight-Through




```
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(confiip address 192.168.2.1interface gip address 192.168.2.1 255.255.255.0ip address
192.168.2.1 255.255.255.0
% 192.168.2.0 overlaps with GigabitEthernet0/0
Router(config-if)#ip address 192.168.ip address 192.168.2.1 255.255.255.0ip address
192.168.2.1intno shutdownno shutdown
% 192.168.2.0 overlaps with GigabitEthernet0/0
GigabitEthernet0/1: incorrect IP aip address 192.168.2.1iip address 192.168.2.1interface gip
address 192.168.2.1 255.255.255.0ip address 192.168.2.1 255.255.255.0
% 192.168.2.0 overlaps with Gip address 192.168.2.1 25no shutdownno shutdown
% 192.168.2.0 overlaps with GigabitEthernet0/0
GigabitEthernet0/1: incorrect IP address assignment
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#eneenable
Router#
Router#
Router#
Router#
Router#enable
Router#cocexit
```

Practical No - 3

AIM - Routing protocal - RIP, EIGRP, OSPR

Problem statement → Set up a multi-router environment and configure dynamic routing protocal (RIP, EIGRP and OSPF) compare their behaviour in terms of convergence, metrics and route selection. Verify that all routers have consistent routing tables and that end-to-end connectivity is achieved.

Theory - Routing is the process of finding the best path for data to travel from one network to another through router. Routers use routing tables to device where to send data packets.

- What is routing protocal

A routing protocal is a set of rules that use to share network information with other router. Automatically learn router, select the best path to reach a destination dynamic routing protocal update route automatically when network changes area.

- Types of routing protocal

① distance vector - RIP
② advanced distance vector (hybrid) EIGRP
③ Link state - OSPR

- RIP (routing information protocal) – RIP is a distance vector routing protocal that we hop count as it metric maximum hop count : 15 (16 means unreach--able). Sends routing updates every 30 seconds simple and easy to configure.

metric used – hop count (no. of router crossed)

advantages – ① every easy to understand
② suitable for small network

disadvantages – ① Not suitable for large network
② slow convergence
③ limited hop count

- EIGRP (enhanced interior gateway routing protocal) – EIGRP is an advanced distance vector routing protocal developed by cisco.

key points – Uses DUAL algorithm to find best path faster convergence than RIP. Support large network uses multiple metrics.

metric used – Bandwidth, delay, reliability load (mostly bandwidth & delay)

Advantages - very fast convergence, efficient use of bandwidth, supports unequal load balancing

disadvantages - cisco proprietary (most cisco devices)

- OSPF (open shortest path first) -
OSPF is a link state routing protocal that uses cost as its metrics

key point - Uses Dijkitra's shortest path algorithm sends update only when changes occure
divides network into areas very fast convergence.

metric used - cost (based on bandwidth)

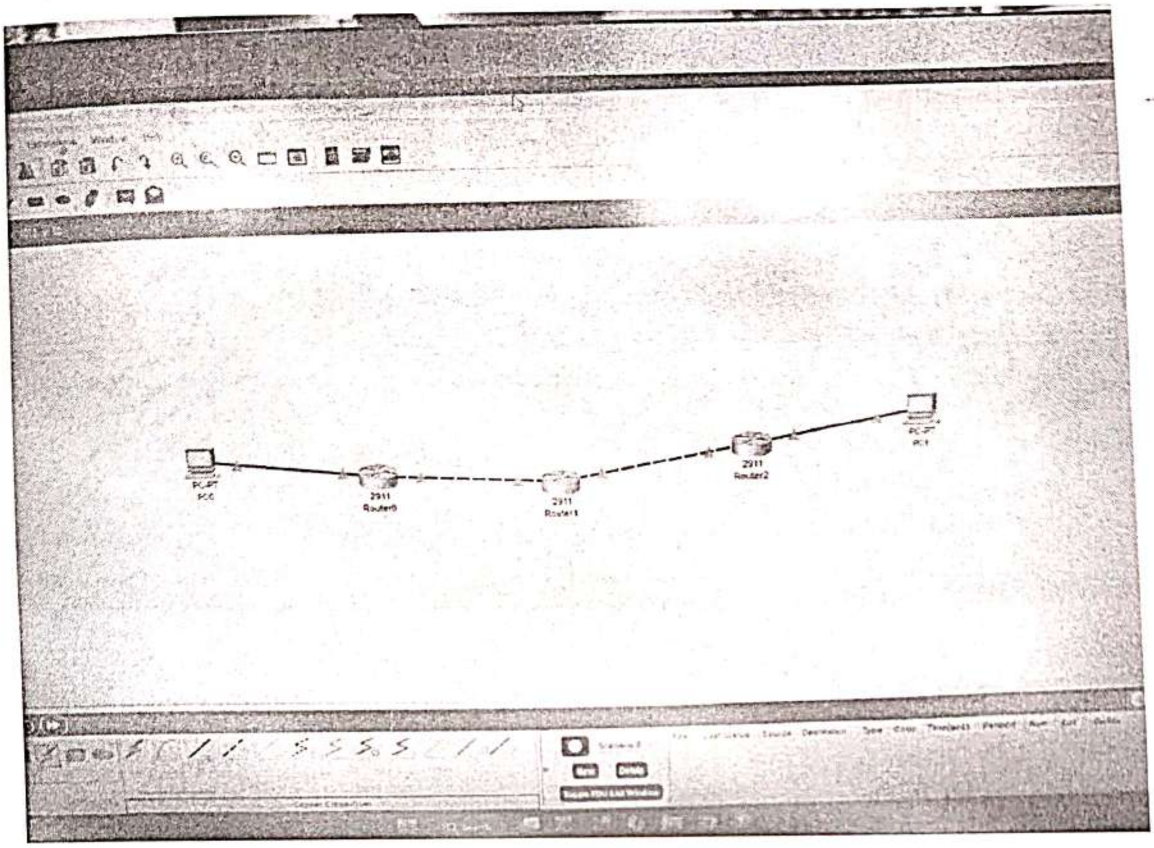
advantage - scable and reliable, fast convergence suitable for large network

disadvantages - more complex configuration require more memory & CPU

Conclusion - Hence we studied the routing protocal RIP, EIGRP, OSPF

29/1/26

CY - Prac - 3



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#exit
```

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
```

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#interface g0/0
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#exit
```

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
C       10.0.0.4/30 is directly connected, GigabitEthernet0/0
L       10.0.0.5/32 is directly connected, GigabitEthernet0/0

Router>
```

Practical No - 4

Aim - Implement IPsec VPN with advanced configuratn.

Theory - IPsec (Internet protocal security) - It is a suits of protocal used to secure IP communicatn by providing authentication, integrity, confidentiality & anti-replay protection at the network layer.

Objective of IPsec

- confidentiality - encrypt data
- Integrity - prevent's data modification
- Authentication - verifies sender identity
- Anti-replay = prevents replay attacks

IPsec protocal. VPNs is used to create a secure tunnel between two endpoints (hosts or gateways) over the internet. This tunnel ensures that sensitive data transmitted between private networks remains confidential & protected from unauthorized access

IPsec works by establishing a set of rules called securely association (SA's). These 'SA' define the security parameters such as encryption algorithm, integrity method, and key management technique used for secure communication.

IKE phase I establishes a secure and authenticated channel between the VPN peers.

IKE phase II create IPsec SA that are used to protect actual data traffic.

IPsec protocal used

- Authentication Header (AH)
provide data integrity and authentication but does not provide encryption

- Encapsulating security protocal (ESP) -
provides encryption, authentication, integrity & anti-replay protection. ESP is commonly used in practical IPsec VPN implementation.

ARISE & SHINE

Modes of operation

- Transport mode - Encrypts only the ~~protocol~~ payload of the IP packet and is used mainly for host to host communication.

- Tunnel mode - Encrypts the entire IP packet and encapsulates it within a new IP header. This mode is widely used in security features

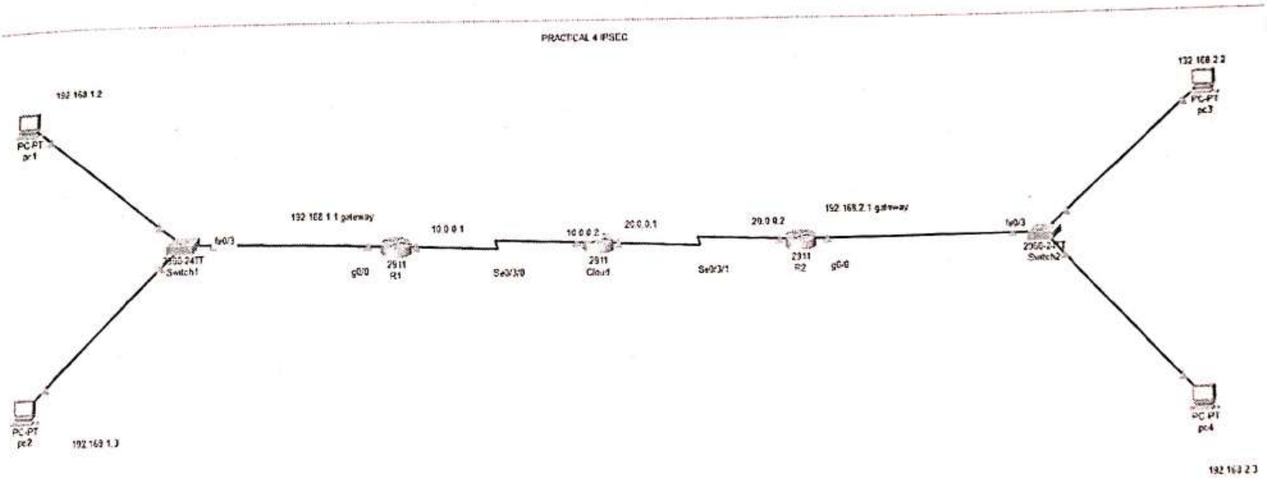
- Encryption ensures confidentiality of data
- Authentication verifies the identity of communicating peers.
- Integrity checking ensures data is not allowed during transmission.
- Anti-replay protection prevents attackers for ensuring captured packets.

Application

- IPsec VPN is used in networking
- connect two branch office's security
- Enable remote users to access corporates networks
- Protect sensitive data over public networks.

Conclusion - IPsec VPN practically provides a secure mechanism for transmitting data over an insecure network by using encryption, authentication and key management techniques making it suitable for enterprise - level secure communication.

Cy- Prac - 4



R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#



Router con0 is now available
```

Copy Paste

Top

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
        --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#
Router(config)#interface serial 0/3/1
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 20.0.0.0
Router(config-router)#network 192.168.3.0
Router(config-router)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
```

Cloud

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#no shutdown
                                  ^
% Invalid input detected at '^' marker.

Router(config-if)#exit
Router(config)#interface serial 0/3/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
exit
Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#network 20.0.0.0
Router(config-router)#
```

Practical No - 5

~~Aim - [illegible]~~

Aim - Wireless security simulation.

Problem statement - Deploy a wireless LAN & implement various wireless security mechanisms such as WPA2 / WPA3, MAC filtering and SSID management. Simulate common wireless & configure counters measures to secure the WLAN. Evaluate the impact of different security levels or connectivity & performance.

Theory - Wireless LAN - A wireless local area network allows devices to connect a network using radio waves instead of physical cables WLAN provides mobility & flexibility but are vulnerable to security threats such as unauthorized access waveshapping & attacks.

Wireless security - Wireless security involves techniques used to protect wireless networks from unauthorized users & attacks. Security mechanisms ensure confidentiality, integrity, & availability of data.

Wireless security mechanism - ① WPA2 (Wifi protocal access)
- uses AES encryption

- stronger than WEP & WPA.

② WPA3
- latest wireless security standard
- provides stronger encryption & protection
- improved authentication mechanisms

③ MAC address filtering
- ~~~~ Allows only specific device from connecting
- parents unauthorized devices from connecting
- can be by passed if MAC spoofing is used.

④ SSID management
- SSID (service set Identifier) is the wireless network name
- Hiding SSID reduces visibility to caused attackers.
- Helps prevent unauthorised access.

Procedure -
ARISE & SHINE

① configure wireless LAN
- Place a wireless Router /access point
- Set SSID name : secure WLAN
- Enable DHCP on the router.

② configure WPA2 security
- select WPA2 - PSK
- encryption : AES
- Password : wireless @ 123.

- connect authorised wireless PC.

③ configure WPA3 security
- change security mode to WPA3
- set strong password
- Reconnect wireless devices.

④ configure MAC address filtering
- Enable MAC filtering on the router
- Add MAC address of authorised PC
- Attempt connecting from unauthentic device

⑤ SSID Management
- Disable SSID broadcast
- Manually configure SSID on client device

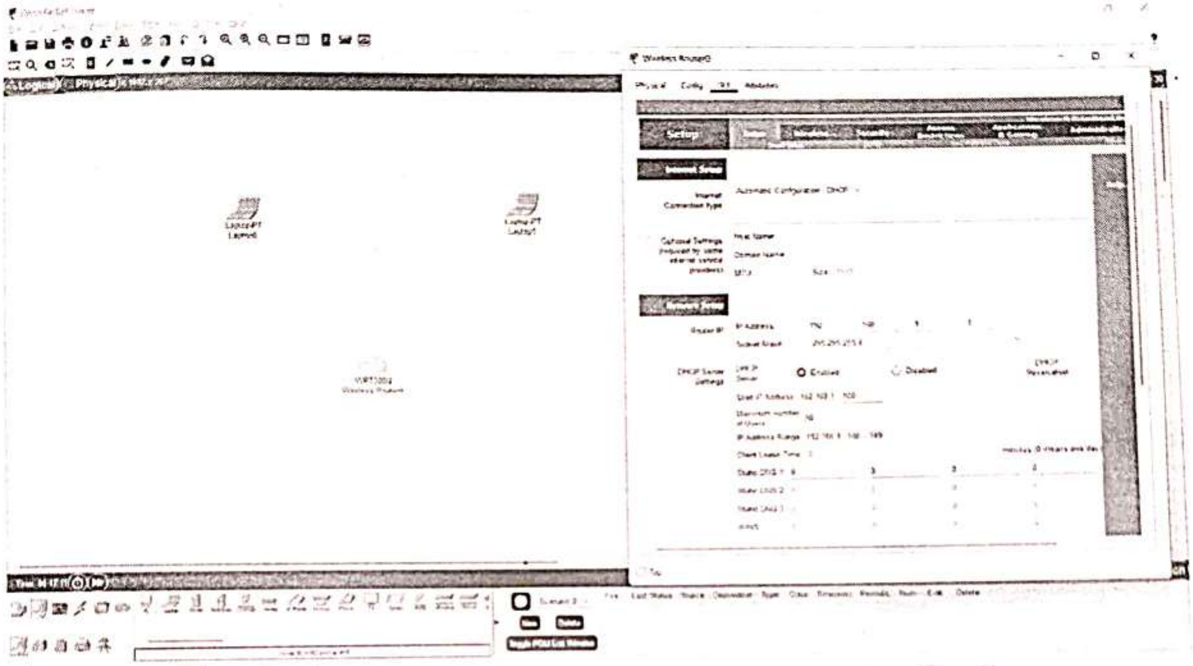
ARISE & SHINE

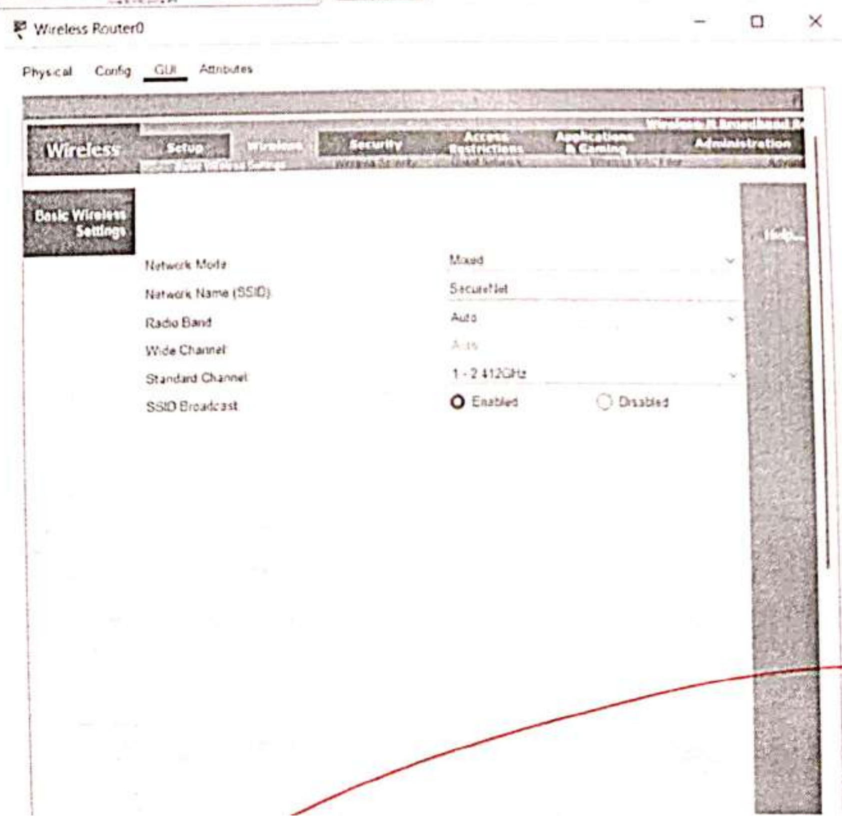
output -
commands → wireless PC connection status
→ Router security setting
→ Packet Tracker simulation results

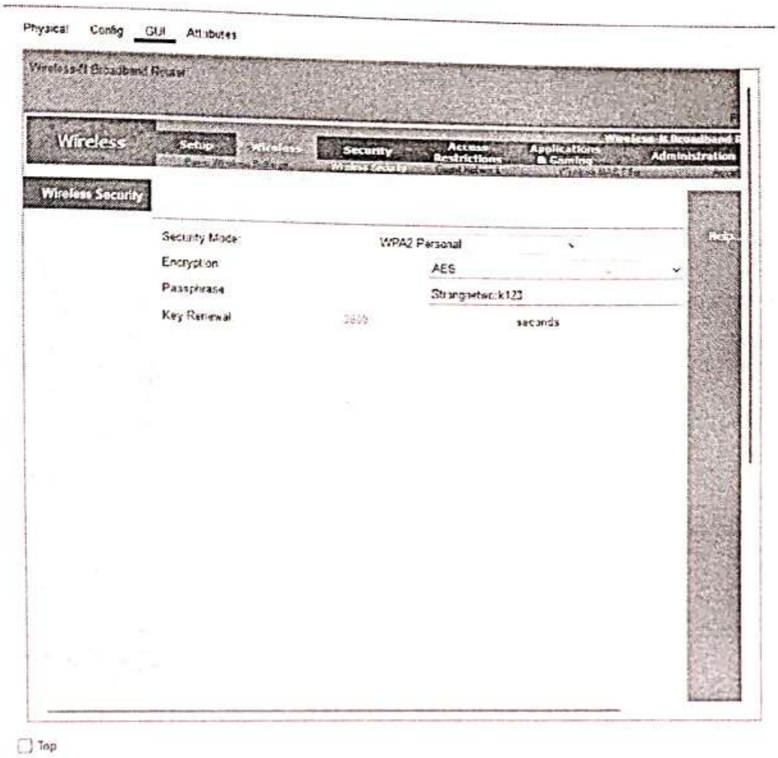
conclusion - This experiment demonstrated the importance of wireless security mechanisms in protecting WLANs. Higher security levels improved protecting against attacks with minimal impact on performance. WPA3 provided the higher level of security.

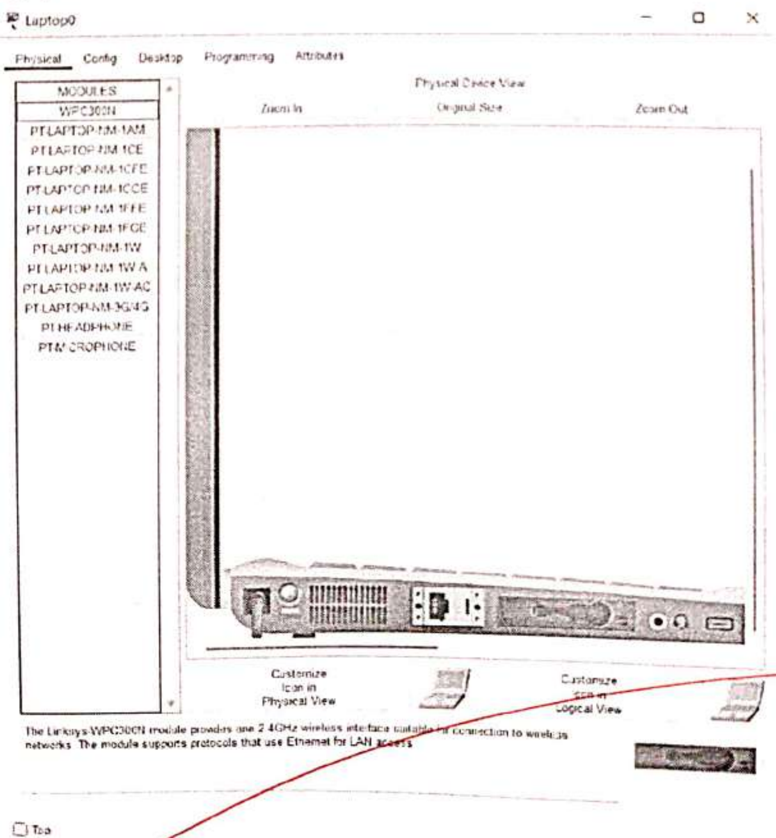
4- Prac-5

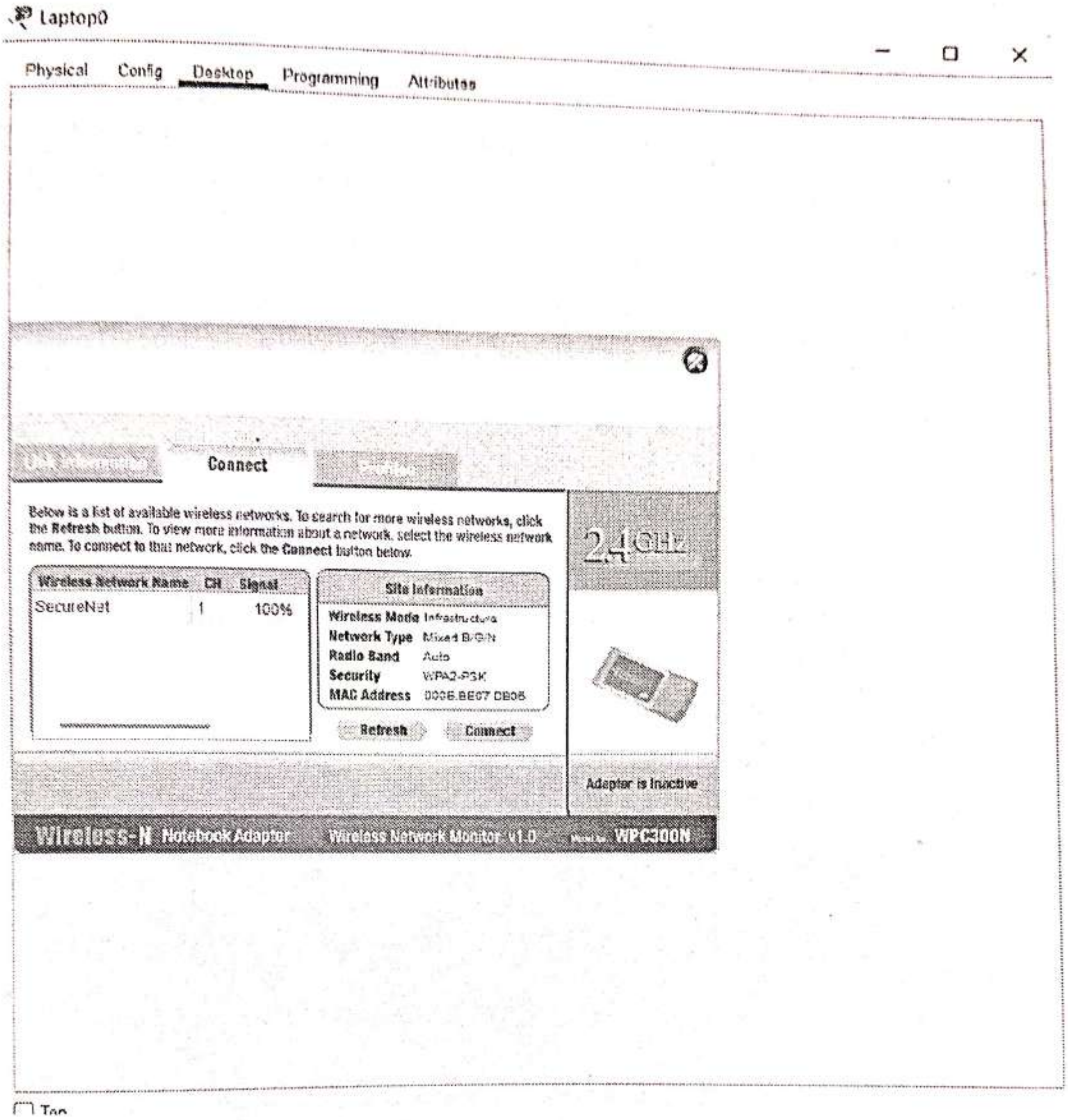
WIRELESS SECURE NETWORK



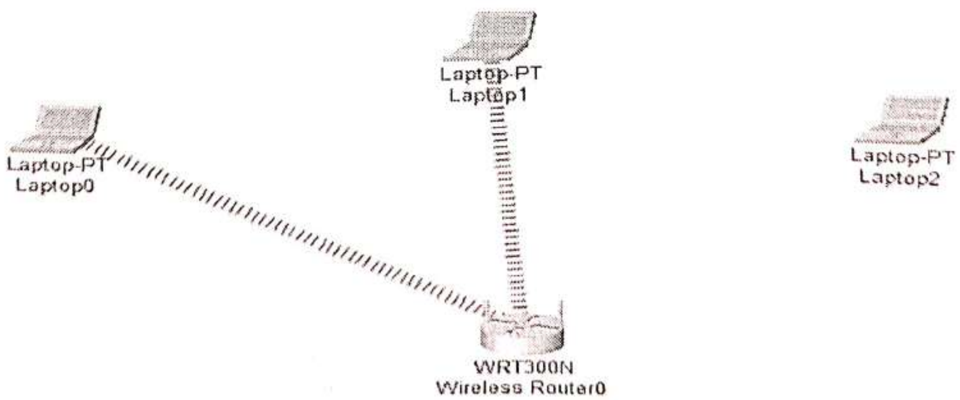



Physical
Config
GUI
Attributes
Wireless
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Wireless Security
Security Mode:
WPA2 Personal
Encryption
AES
Passphrase
Strongnetwork123
Key Renewal
seconds
Top

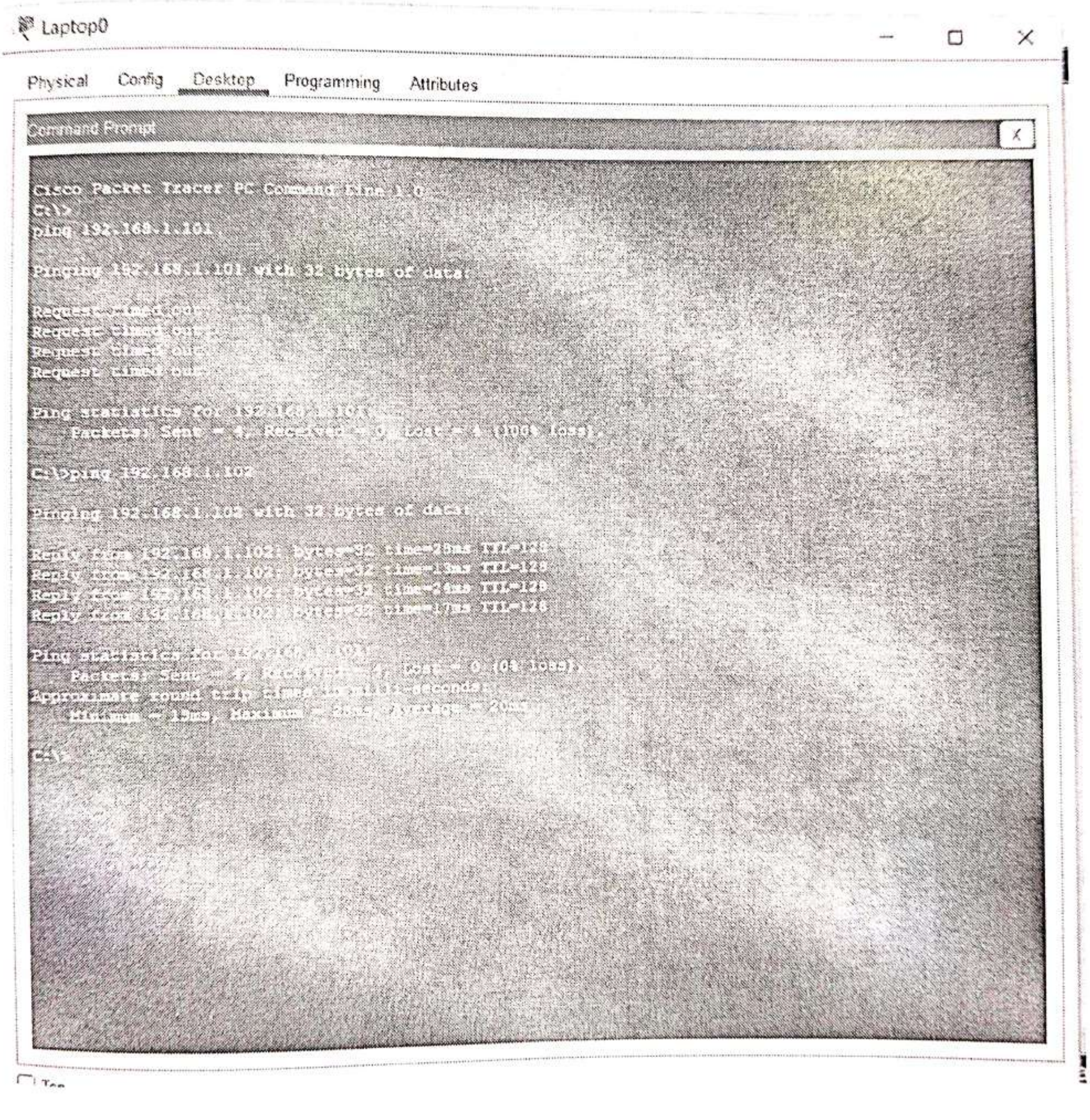

Laptop0
Physical
Config
Desktop
Programming
Attributes
MODULES
WPC300N
PT-LAPTOP-NM-1AM
PT-LAPTOP-NM-1CE
PT-LAPTOP-NM-1CFE
PT-LAPTOP-NM-1CGE
PT-LAPTOP-NM-1FFE
PT-LAPTOP-NM-1FGE
PT-LAPTOP-NM-1W
PT-LAPTOP-NM-1W-A
PT-LAPTOP-NM-1W-AC
PT-LAPTOP-NM-3G/4G
PT-HEADPHONE
PT-MICROPHONE
Physical Device View
Zoom In
Original Size
Zoom Out
Customize Icon in Physical View
Customize Icon in Logical View
The Linksys-WPC300N module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.
Top



Laptop0
Physical
Config
Desktop
Programming
Attributes
Connect
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
CH
Signal
SecureNet
1
100%
Site Information
Wireless Mode
Infrastructure
Network Type
Mixed B/G/N
Radio Band
Auto
Security
WPA2-PSK
MAC Address
Refresh
Connect
2.4GHz
Adapter is Inactive
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
WPC300N



Laptop-PT
Laptop1
Laptop-PT
Laptop0
Laptop-PT
Laptop2
WRT300N
Wireless Router0


Laptop0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.1.101
Pinging 192.168.1.101 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.101:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.102
Pinging 192.168.1.102 with 32 bytes of data:
Reply from 192.168.1.102: bytes=32 time=25ms TTL=128
Reply from 192.168.1.102: bytes=32 time=13ms TTL=128
Reply from 192.168.1.102: bytes=32 time=24ms TTL=128
Reply from 192.168.1.102: bytes=32 time=17ms TTL=128
Ping statistics for 192.168.1.102:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 13ms,
C:\>

Practical NO - 6

Aim - Multilayer switch configuration

Problem statement - Configure a multilayer switch to perform both layer -2 switching a layer - 3 routing. Implement VLANs, SVI, routing protocal & port security. Demonstrate high performance inter - VLAN communication & verify connects routing & switching operations

Theory - A multilayer switch is a network device that perform ~~bas~~ both layer-2 switching & layer-3 routing functions unlike traditional switches, multilayer switches can route packets between VLANs at hardware speed using switch virtual Interface (SVI). This provides high performance inter VLAN communication.

ARISE & SHINE

Network topology — PC1 (VLAN10) — Multilayer switch
PC 2 (VLAN20) — Multilayer switch
PC3 (VLAN 30)

IP Addressing scheme -

VLAN	Network	SVI IP
10 (PC1)	192.168.10.0/24	192.168.10.1
20 (PC2)	192.168.20.0/24	192.168.20.1
30 (PC3)	192.168.30.0/24	192.168.30.1

configuration -

1) Enable Routing on switch
- enable
- configure terminal
#ip routing

2) Create VLANs
#VLAN 10
name sales
VLAN 20
name HR
VLAN 30
name IT

3) Assign ports to VLANs :
interface fastEthernet 0/1
switchport mode access
switchport access VLAN 10

#interface fastEthernet 0/2
switchport mode access
switchport access VLAN20

interface fastEthernet 0/3
#switchport mode access
#switchport access VLAN 30

4) configure SVIs :
interface VLAN 10
ip address 192.168.10.1 255.255.255.0
no shutdown

interface VLAN 20
ip address 192.168.20.1 255.255.255.0
no shutdown

interface VLAN30
ip address 192.168.30.1 255.255.255.0
no shutdown

5) configure port security (PCI port)
interface fastEthernet 0/1
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
switchport port-security mac-address sticky

verification
show VLAN brief
show ip route
show port-security interface fastEthernet 0/1.

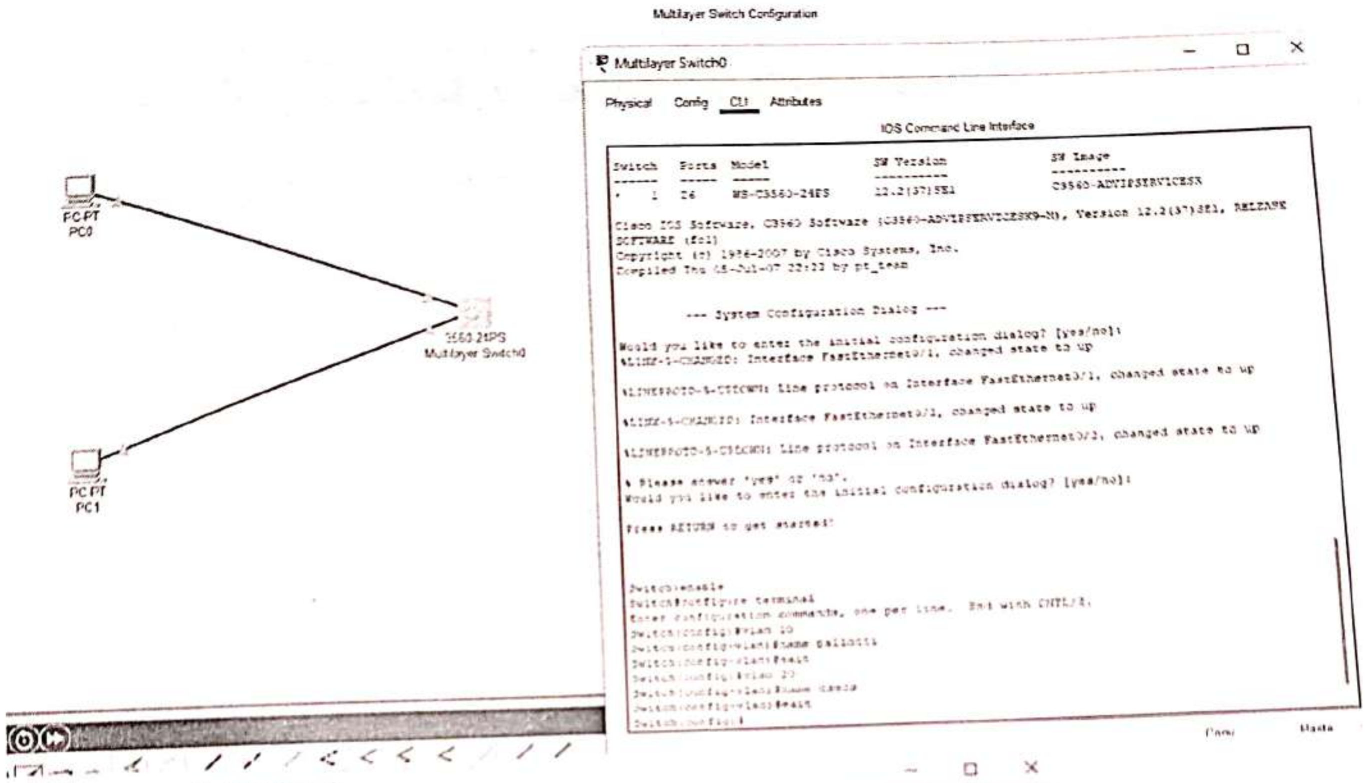
Testing -

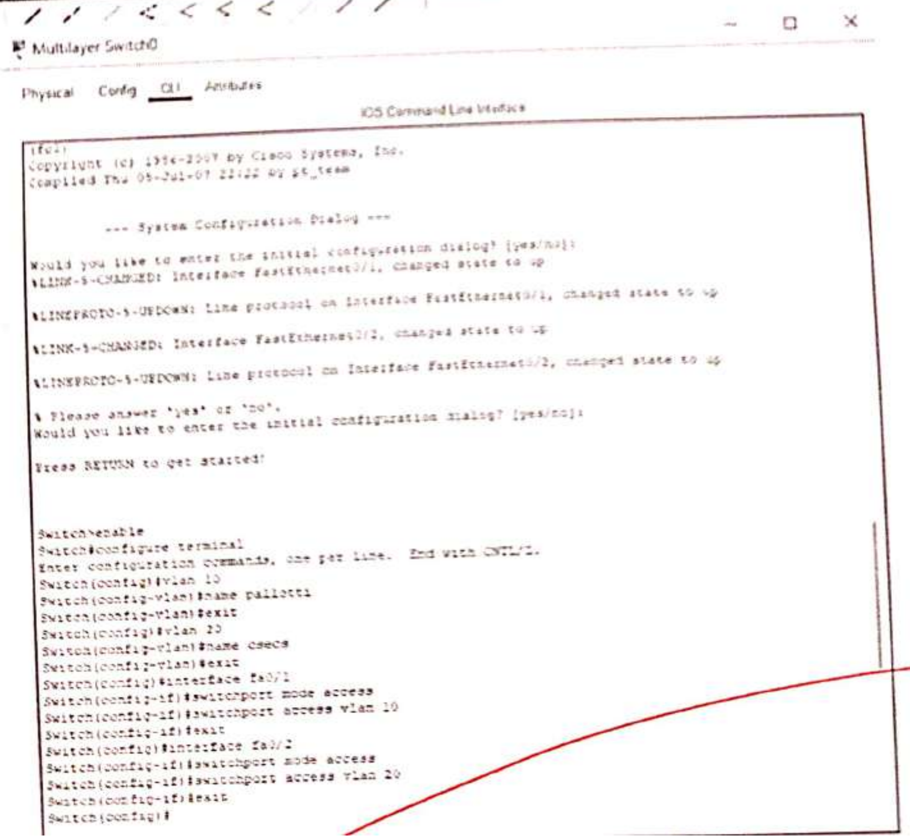
ping PC1 → PC2 / PC3

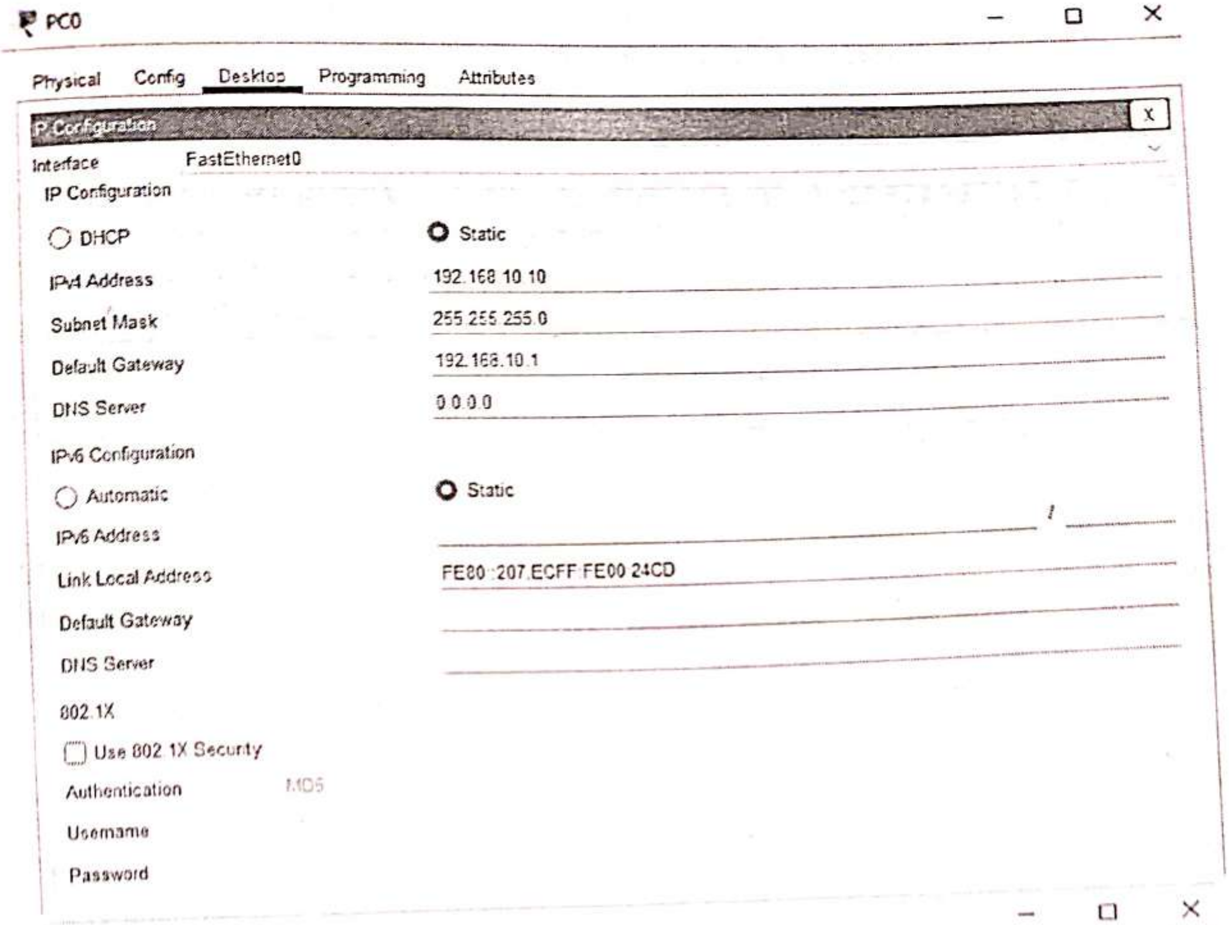
ping PC2 → PC3

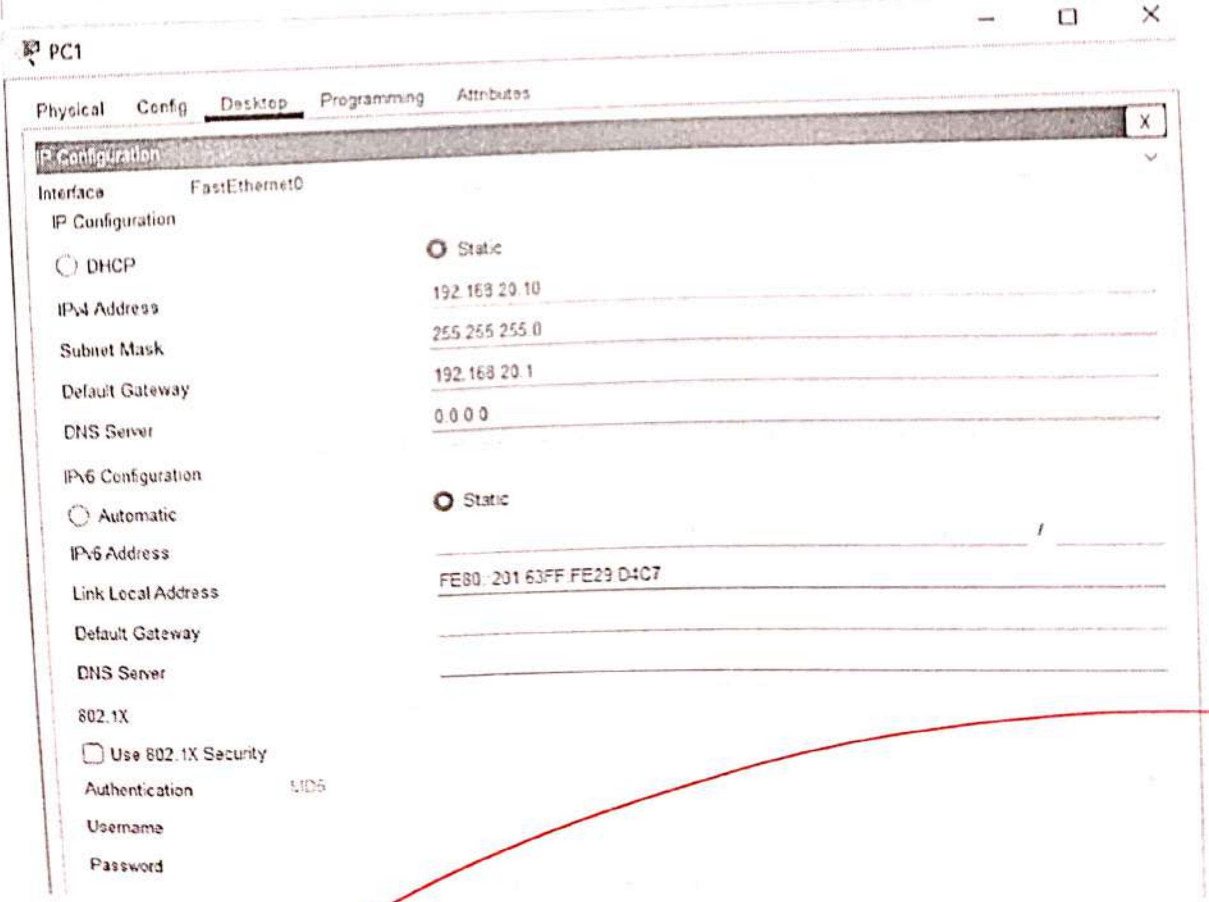
conclusion - This experiment demonstrate hae a multilayer switch LAN replace traditional router - or - a static architecture by providing efficient inter VLAN routing & enhanced security using port security, resulting in a high performance & secure network.

ARISE & SHINE

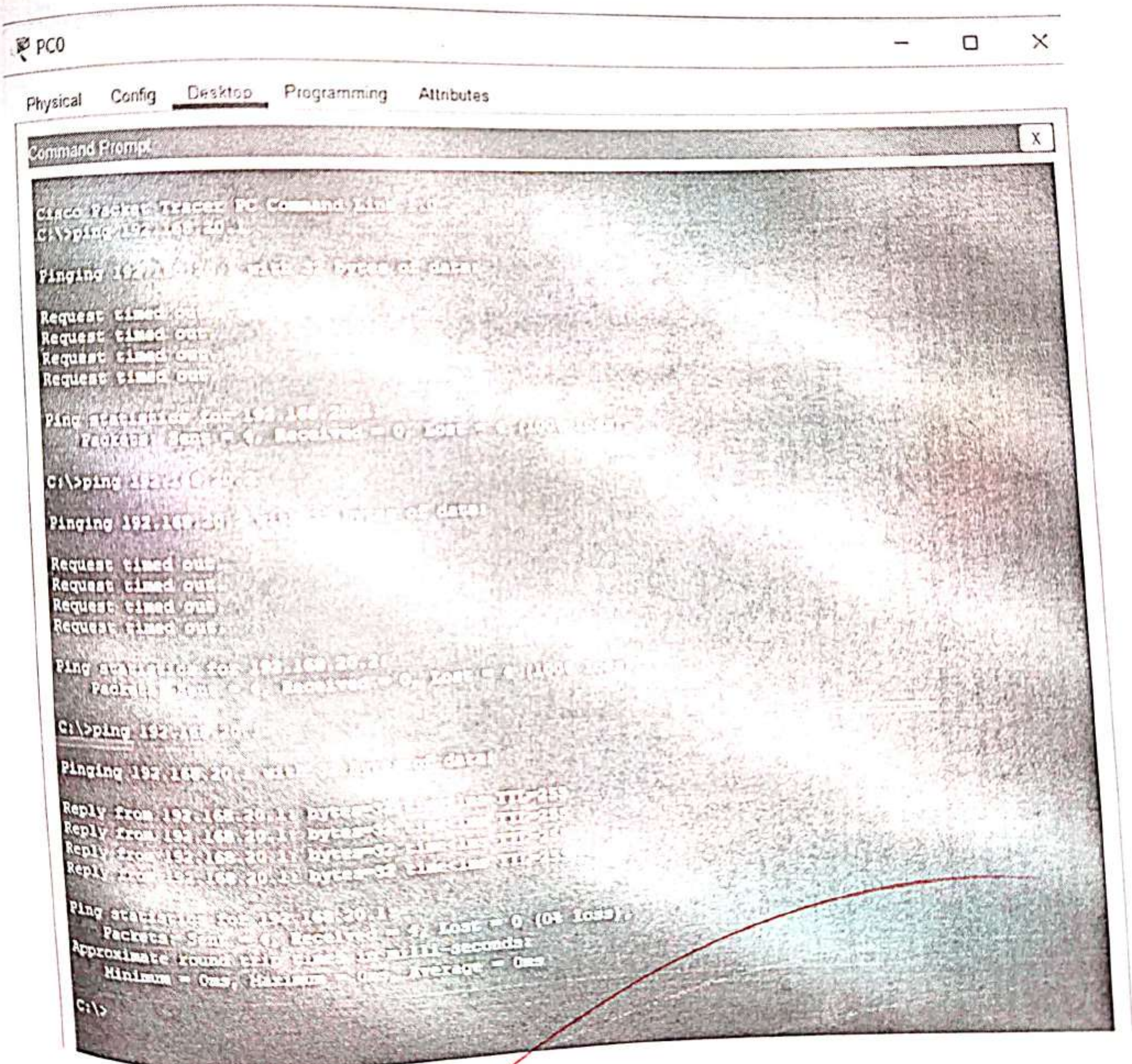

Multilayer Switch Configuration
PC PT
PC0
3560 24PS
Multilayer Switch0
PC PT
PC1
Multilayer Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch Ports Model SW Version SW Image
------ ----- ----- ---------- ----------
* 1 26 WS-C3560-24PS 12.2(37)SE1 C3560-ADVIPSERVICESK
Cisco IOS Software, C3560 Software (C3560-ADVIPSERVICESK9-M), Version 12.2(37)SE1, RELEASE SOFTWARE (fc1)
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Thu 05-Jul-07 22:22 by pt_team
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name pallotti
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name cseca
Switch(config-vlan)#exit
Switch(config)#
Copy
Paste


Multilayer Switch0
Physical Config CLI Attributes
IOS Command Line Interface
(fc1)
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Thu 05-Jul-07 22:22 by pt_team
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name pallotti
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name cseca
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#

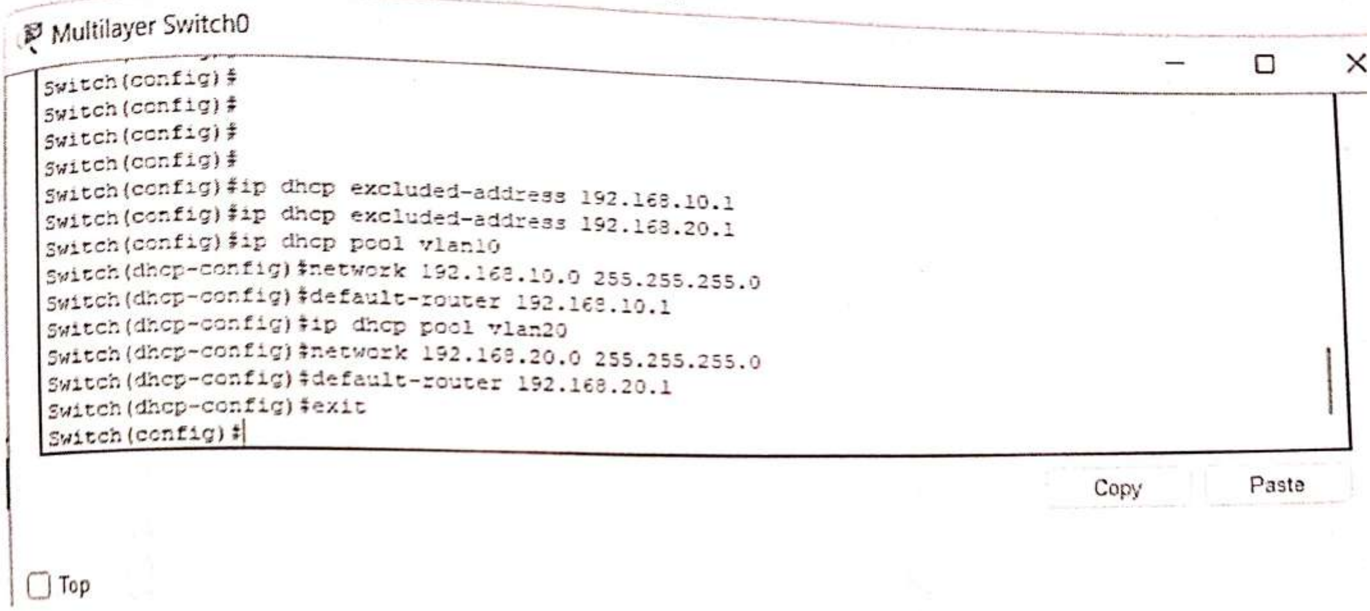

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::207:ECFF:FE00:24CD
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

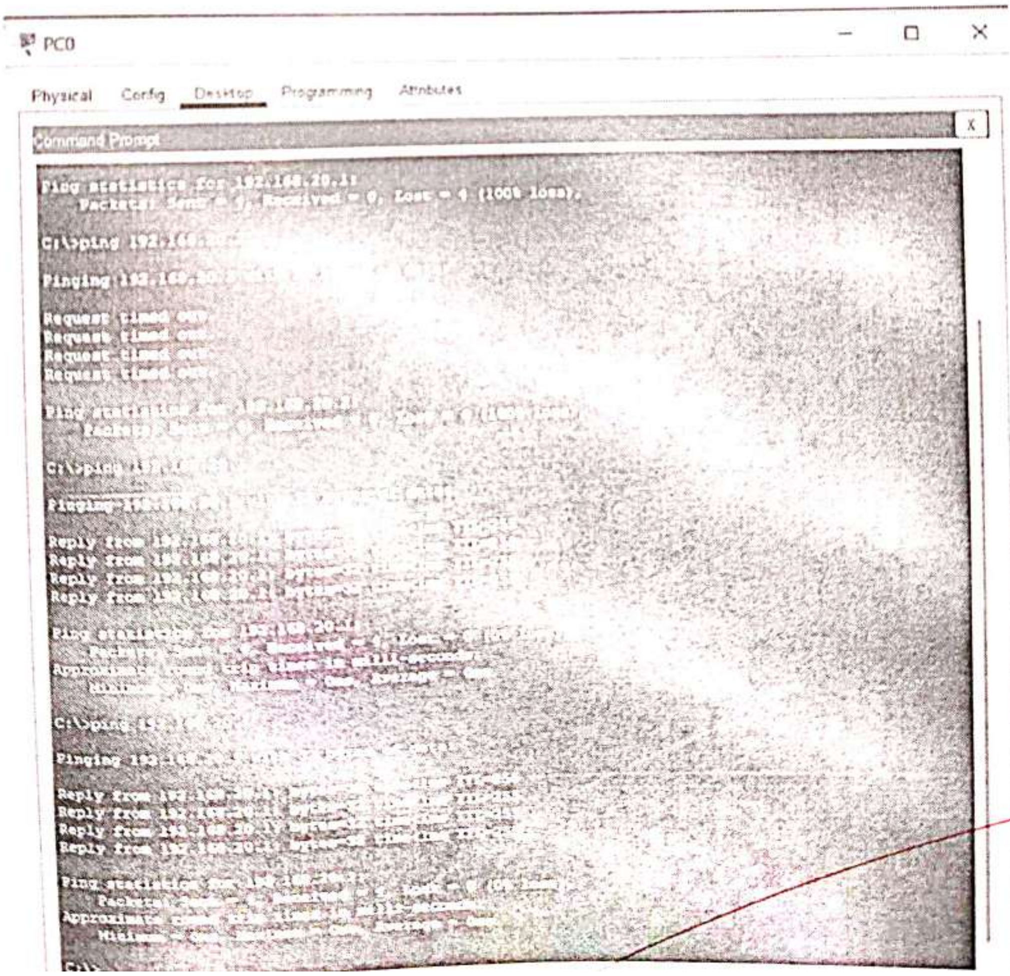

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:63FF:FE29:D4C7
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

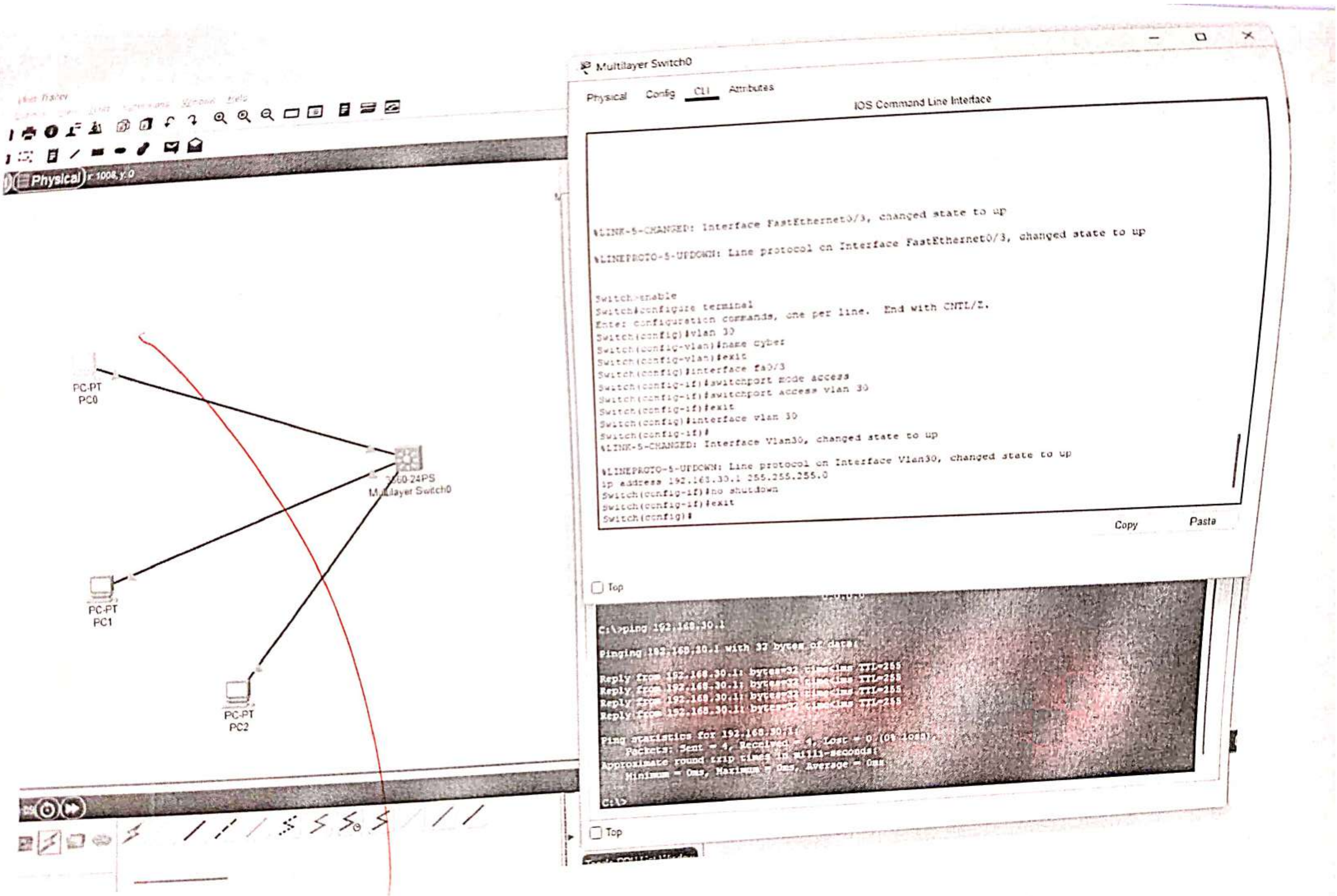
```
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```



DHCP


Multilayer Switch0
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#ip dhcp excluded-address 192.168.10.1
Switch(config)#ip dhcp excluded-address 192.168.20.1
Switch(config)#ip dhcp pool vlan10
Switch(dhcp-config)#network 192.168.10.0 255.255.255.0
Switch(dhcp-config)#default-router 192.168.10.1
Switch(dhcp-config)#ip dhcp pool vlan20
Switch(dhcp-config)#network 192.168.20.0 255.255.255.0
Switch(dhcp-config)#default-router 192.168.20.1
Switch(dhcp-config)#exit
Switch(config)#
Copy
Paste
Top


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Ping statistics for 192.168.20.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),


Multilayer Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#name cyber
Switch(config-vlan)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface vlan 30
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
ip address 192.168.30.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Copy
Paste
Top
C:\>ping 192.168.30.1
Pinging 192.168.30.1 with 32 bytes of data:
Ping statistics for 192.168.30.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Physical
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
3560-24PS
Multilayer Switch0

Practical No — 7

Aim - IPv6 network setup.

Problem statements - Design an IPv6 based network addressing LAN and configure IPv6 addressing on router and hosts. Ensure proper communication using link-local, global unicast, and auto configuration methods (SLAAC).
Test IPv6 connectivity using ICMPv6 and neighbour discovering tools.

Theory - IPv6 (Internet protocal version 6) is the latest version of IP address system used to identify devices on a network
- IPv4 example : 192.168.1.1
- IPv6 example : 200:0db8:8503:0000:0000:802e:0370:7334

IPv6 is created because IPv4 addresses ~~were~~ were getting finished.

• **Why do we need IPv6?**
- Huge number of addresses
- Better security
- Faster routing
- Built-in auto configuration

• Types of IPv6 addresses

① Link-local addresses
- Start with FE80 ::
- used for communication inside same network
- Automatically created
- Cannot used on internet
eg - FE80 ::1

② Global unicast address
- used for internet communication
- public address
- similar to public IPv4
eg - 2001:DB8::1

③ Unique local address
- similar to private IPv4
- starts with FC00 :: or FD00 ::

• IPv6 network addressing plan (design)
- 1 Router
- 2 LAN network
- 4PCs

LAN 1 → 2001 : DB8 : 1 :: 1 64 → 2001 : DB8 : 1 :: 1
LAN 2 → 2001 : DB8 : 2 :: 1 64 → 2001 : DB8 : 2 :: 1

- configure IPv6 on PC

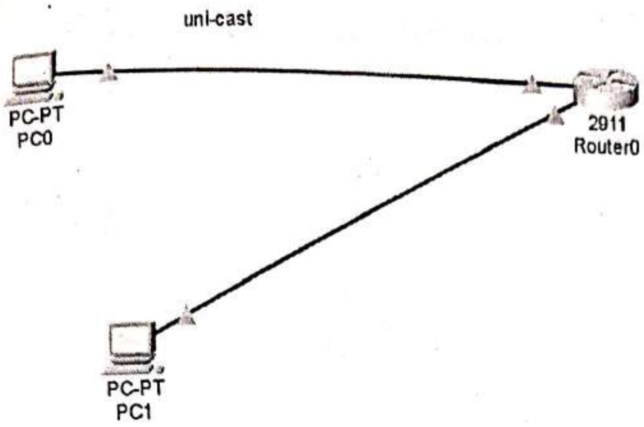
manual configuration → enter IPv6 address
- Enter prefix /64
- Enter default gateway.

method 2 : SLAAC (Auto configuration)
SLAAC - stateless address Auto configuration.
- Router automatically sends address info.
- PC automatically generates its own IPv6 address
- No DHCP server needed.
- Router sends router advertisement (RA) message.

- PC combines :
- Network prefix (from router)
- Interface ID (from MAC address)
then creates full IPv6 address automatically.

ARISE & SHINE

Conclusion - Hence, we studied IPv6 network setup, is the latest internet protocal that overcome the limitations of IPv4 by providing a very large address space.



PC0

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address:

Subnet Mask:

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

Automatic | Static

IPv6 Address: 2001:DB8:1:1::10 / 64

Link Local Address: FE80::20D:BDFF:FE2E:6D8C

Default Gateway: 2001:DB8:1:1::1

DNS Server:

802.1X

Use 802.1X Security

Authentication: MD5

PC1

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address:

Subnet Mask:

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

Automatic | Static

IPv6 Address: 2001:DB8:1:2::10 / 64

Link Local Address: FE80::20A:F3FF:FE8D:3291

Default Gateway: 2001:DB8:1:2::1

DNS Server:

802.1X

Use 802.1X Security

Authentication: MD5

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface g0/0
Router(config-if)#ipv6 address 2001:DB8:1:1::1/64
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
exit
Router(config)#interface g0/1
Router(config-if)#
Router(config-if)#ipv6 address 2001:DB8:1:2::1/64
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up
exit
Router(config)#
```

```
C:\>
C:\>ping 2001:DB8:1:2::10

Pinging 2001:DB8:1:2::10 with 32 bytes of data:

Reply from 2001:DB8:1:2::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::10: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:1:2::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping FF02::1

Pinging FF02::1 with 32 bytes of data:

Reply from FE80::240:BFF:FEEE:BA01: bytes=32 time<1ms TTL=255
Reply from FE80::240:BFF:FEEE:BA01: bytes=32 time<1ms TTL=255
Reply from FE80::240:BFF:FEEE:BA01: bytes=32 time<1ms TTL=255
Reply from FE80::240:BFF:FEEE:BA01: bytes=32 time<1ms TTL=255

Ping statistics for FF02::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

Practical No - 8

Aim - IPv6 Routing configuration.

Problem statement - Implement dynamic IPv6 routing protocals such as RIPng, OSPF v3 or EIGRP for IPv6 in a multi-router environment. Ensure that all routers envirnoment. correctly and achieves full network reachability. Validate route propagation and fallover behaviour

Theory - IPv6 routing means sending data (packets) from one network to another using IPv6 address

- Just like google maps shows route between places, routers find the best path for data.

- Router use routing protocals to share information with each other.

ARISE & SHINE

• Why do we need dynamic routing:
There are two types of routing

① Static Routing
- You manually enter routers
- works for small network

② Dynamic Routing
- Routers automatically learn routes
- If network changes → routers adjust automatically.
- saves times and reduces errors

• Main IPv6 routing protocals

① RIPng (Routing Information protocal next generation)

Features - uses hop count (number of routers to find best path.
- maximum 15 hops allowed.
- slow covergence (takes time to update changes)

Best for → small networks

ARISE & SHINE

eg - If routers A → router B → router C
hop count = 2 → shortest path chosen

② OSPFv3 (open shortest path first version 3)

Features → uses cost (bandwidth) instead of hop count

- Faster than RIPng
- supports large networks

- uses link - state algorithm.

Advantage - very fast convergence
- efficient routing

Best for - medium to large ~~network~~ network

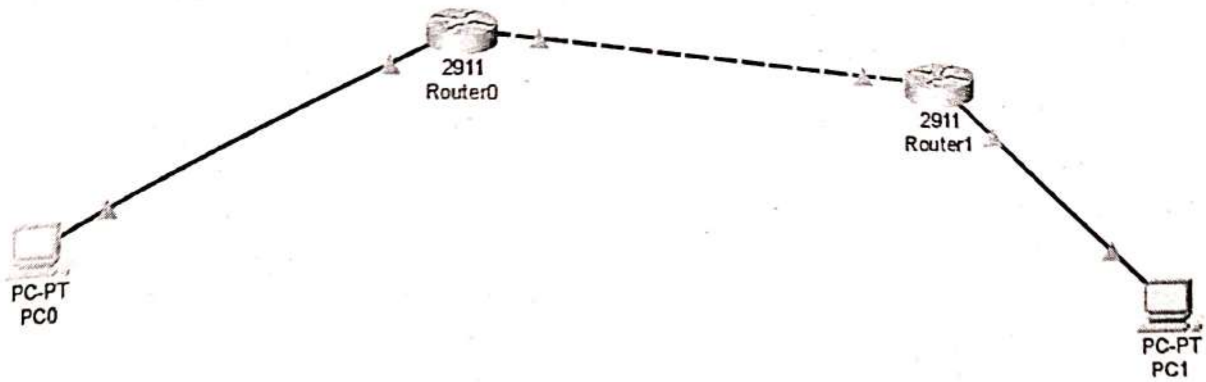
- Validation of Routing
To check if everything is working.

use commands

- ping → check connectivity
- Traceworks → check path
- short IPv6 route → check routing table

conclusion - Hence, we studied IPv6 routing configuration

22/4/26




```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface g0/0
Router(config-if)#ipv6 address 2001:DB8:1::1/64
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up
exit
Router(config)#interface g0/1
Router(config-if)#ipv6 address 2001:DB8:12::1/64
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#ipv6 route 2001:DB8:2::/64 2001:DB8:12::2
Router(config)#
```

Copy Paste

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface g0/1
Router(config-if)#ipv6 address 2001:DB8:12::2/64
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up
exit
Router(config)#interface g0/0
Router(config-if)#ipv6 address 2001:DB8:2::1/64
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up
exit
Router(config)#ipv6 route 2001:DB8:1::/64 2001:DB8:12::1
Router(config)#
```

Copy Paste

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping  2001:DB8:2::10

Pinging 2001:DB8:2::10 with 32 bytes of data:

Reply from 2001:DB8:2::10: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:2::10: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:2::10: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:2::10: bytes=32 time<1ms TTL=126

Ping statistics for 2001:DB8:2::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Practical No - 9

Aim - A-A-A configuration.

Problem statement - configure authentication, authorization, and accounting (AAA) services using or TACACS + or RADIUS server. Implement centralized login control for network devices. Test user authentication, command authorization and activity logging to ensure secure and control administrative access

Theory - AAA stands for

A → Authentication

A → Authorization

A → Accounting

It is used to control and secure access to network devices like routers and switches:

ARISE & SHINE

① Authentication - (who are you?)

- This steps checks user identify
- users enters username + password
- system verifies if the users is valid or not
- eg - like logging into your phone using password or fingerprint.

② Authorization – (What can you do?)
After login, it decides what permission users has –
(i) which command user can run
(ii) what access level they get
eg - Admin → full control
user → limited access

③ Accounting – (what did you do?)
This steps keeps record of users activity
- When users logged in/out
- what command they executed
- How long they stayed
eg - like history / log of all actions.

• What is centralized AAA :
Instead of storing users on each routers, we use a central server
Two main server –
• TACACS + (Terminal Access controller access control system plus)
• RADIUS (Remote authentication dial-in user service)

P.T.O.

ARISE & SHINE

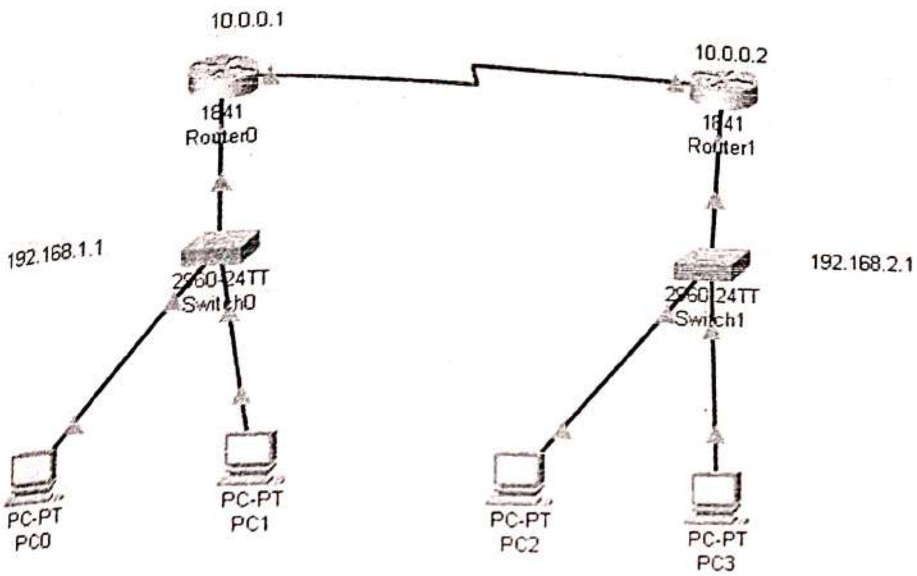
- Purpose of AAA
- Improves security
- provides central control
- Tracks user activities
- Prevents unauthorized access

- How AAA works

1) users tries to login
2) Device sends request to AAA server
3) Servers checks username & password.
4) If correct → access granted
5) server decides permission
6) All activities are recorded

AAA is a security framework used to control access to network devices by verifying user identity (Authentication), giving permission (authorization) and recording activities (accounting)

Conclusion - Hence, we studied AAA configuration




```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Router(config-if)#exit
Router(config)#interface f0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state
to up

Router(config-if)#exit
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#ip dhcp pool cs
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
state to up

Router(config)#interface f0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#exit
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#ip dhcp pool cs
Router(dhcp-config)#network 192.168.2.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.2.1
Router(dhcp-config)
```

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time=7ms TTL=126
Reply from 192.168.2.3: bytes=32 time=11ms TTL=126
Reply from 192.168.2.3: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 11ms, Average = 8ms

C:\>
```

```
Building configuration...
[OK]
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#aaa new-model
Router(config)#username admin password hello
Router(config)#aaa authentication login default local
Router(config)#enable passord cisco
                          ^
% Invalid input detected at '^' marker.

Router(config)#enable password cisco
Router(config)#line console 0
Router(config-line)#login authentication default
Router(config-line)#exit
Router(config)#line vty 0 4
Router(config-line)#login authentication default
Router(config-line)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

```
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#aaa new-model
Router(config)#radius-server host 192.168.2.1 key radius_key
Router(config)#aaa authentication login default group radius local
Router(config)#username backup_admin password backup123
Router(config)#enable  password cisco
Router(config)#line vty 0 4
Router(config-line)#login authentication default
Router(config-line)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

Copy Paste

```
User Access Verification

Username: admin
Password:
Router>
```

```
User Access Verification

Username: backup_admin
Password:
Router>enable
Password:
Router#
```

```
C:\>telnet 10.0.0.1
Trying 10.0.0.1 ...Open


User Access Verification

Username: admin
Password:
Router>^Z
Router>exit

[Connection to 10.0.0.1 closed by foreign host]
C:\>telnet 10.0.0.2
Trying 10.0.0.2 ...Open


User Access Verification

Username: backup_admin
Password:
Router>
```

Practical No - 10

Aim - Bridge connection between routing protocal.

Problem statement - Integrate two or more different routing protocals in a single network and configure route redistribution between them. Ensures seamless communication across domains running protocals such as RIP, EIGRP and OSRF. Validate proper redistribution while avoiding routing loops and suboptimal paths.

Theory - Bridge ~~commun~~ connection between routing protocal : Routing protocal is a set of rules that routers uses to decide the best path to send data from one network to another.

eg - RIP (Routing Information Protocal)
- EIGRP (Enhanced interior gateway routing protocal)
- OSPF (open shortest path first)

ARISE & SHINE

- sometimes, different parts of a network use different routing protocals

eg - one network uses RIP
- another network uses OSPF

These networks cannot understand each other directly.

solution - we use something called route redistribution

It acts like a bridge (connection) between different routing protocals

• What is Route Redistribution?

Route redistribution means sharing routes from one routing protocal to another.

eg - Router from RIP are shared with OSPF
- Router from OSPF are shared with EIGRP
so, all networks can communicate easily

• Why is it needed? → To connect different networks using different protocals.

→ To allow smooth communication (seamless communicatn)
→ To make a large network work as one system.

ARISE & SHINE

• How its works —

i) A router runs multiple routing protocals
ii) It takes routers from one protocal.
iii) Converts them into another protocal format.
iv) Shares them with the other network. this router is called as Redistribution error.

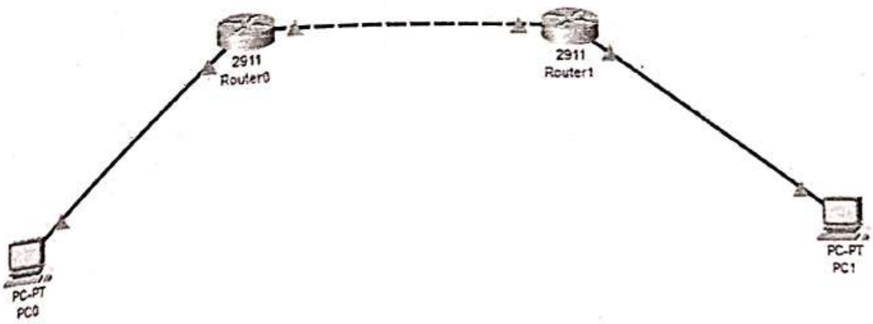
Real life example → RIP, OSPF
Route redistribution = translater
It helps both language understand each other and communicate

- How to avoid problem?
- Uses proper metrics ((cost values)
- Uses route filtering (control which routes are shared)
- Avoid redistributing the same route again and again

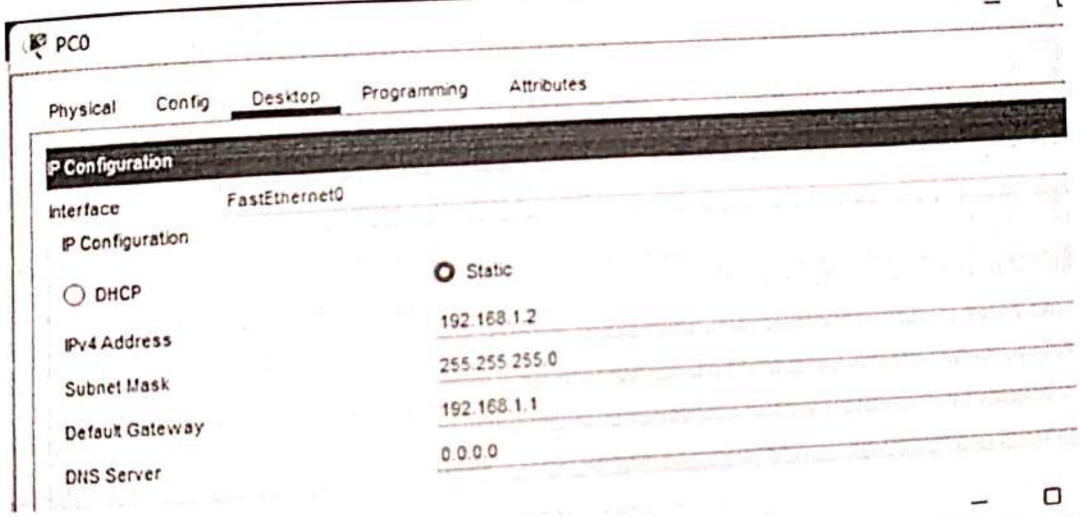
Conclusion - Hence, we studied, bridge connection between routing protocal

ARISE & SHINE

PC – ROUTER CONNECTION



IP ASSIGN:


PC0

Physical Config Desktop Programming Attributes

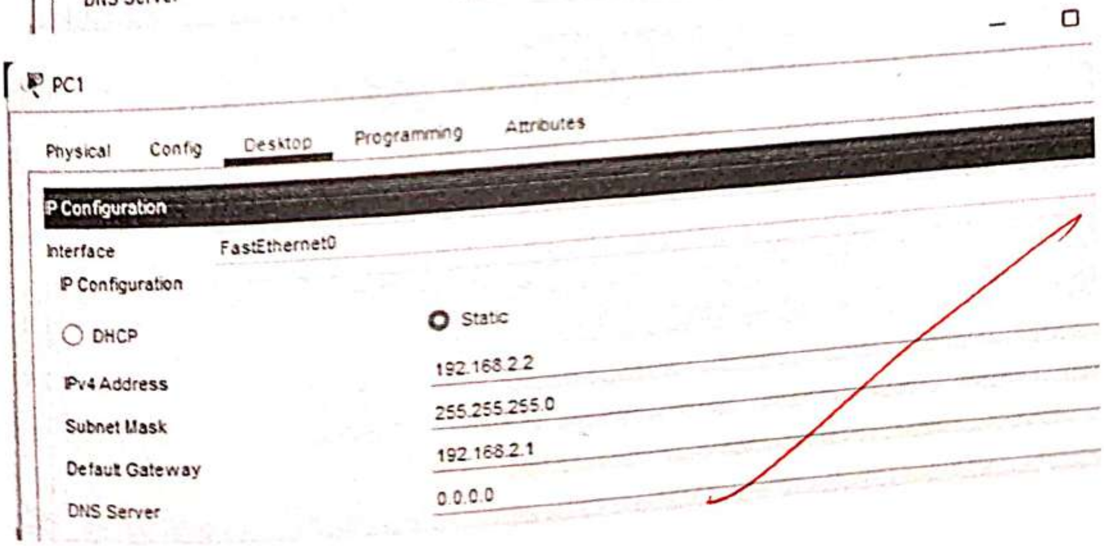
IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0


PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address	192.168.2.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	0.0.0.0

CREATING INTERFACE

ROUTER 0

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0?
/
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

ROUTER 1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

ENABLING RIP : ROUTER 0

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summany
                                  ^
% Invalid input detected at '^' marker.

Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
```

ENABLING OSPF : ROUTER 1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.2.0
Router(config-router)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#router ospf 1
Router(config-router)#redistribute rip subnets
Router(config-router)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#redistribute ospf 1 metric 2
Router(config-router)#exit
Router(config)#
```

PING PC1 FROM PC0

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PING PC0 FROM PC1

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```